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Membrane receptors for antigen

Introduction, 59

The B-cell surface receptor for antigen (BCR), 59

The B-cell inserts a transmembrane immunoglobulin into its surface, 59

The surface immunoglobulin is complexed with associated membrane proteins, 59

The T-cell surface receptor for antigen (TCR), 60

The receptor for antigen is a transmembrane heterodimer, 60

There are two classes of T-cell receptors, 62

The encoding of T-cell receptors is similar to that of immunoglobulins, 62

The CD3 complex is an integral part of the T-cell receptor, 63

The generation of diversity for antigen recognition, 63

Intrachain amplification of diversity, 64

Interchain amplification, 68

Somatic hypermutation, 68

NK receptors, 69

The major histocompatibility complex (MHC), 70

Class I and class II molecules are membrane-bound heterodimers, 70

Several immune response-related genes contribute to the remaining class III region of the MHC, 73

Gene map of the MHC, 73

The genes of the MHC display remarkable polymorphism, 75

Nomenclature, 75

Inheritance of the MHC, 76

The tissue distribution of MHC molecules, 77

MHC function, 77

The nonclassical MHC and class I chain-related molecules, 77

Summary, 78

Further reading, 79

INTRODUCTION

The interaction of lymphocytes with antigen takes place through binding to specialized cell surface antigen-specific receptors functioning as recognition units. Since B- and T-cells use quite distinct receptors to bind antigen, they will have to be discussed separately.

THE B-CELL SURFACE RECEPTOR FOR ANTIGEN (BCR)

The B-cell inserts a transmembrane immunoglobulin into its surface

In Chapter 2 we discussed the cunning system by which an antigen can be led inexorably to its doom by selecting the lymphocytes capable of making antibodies complementary in shape to itself through its ability to combine with a copy of the antibody molecule on the lymphocyte surface. It will be recalled that combination with the surface receptor can activate the cell to proliferate before maturing into a clone of plasma cells secreting antibody specific for the inciting antigen (cf. figure 2.11).

Immunofluorescent staining of live B-cells with labeled anti-immunoglobulin (anti-Ig) (e.g. figure 2.6c)

reveals the earliest membrane Ig to be of the IgM class. The cell is committed to the production of just one antibody specificity and so transcribes its individual rearranged *VJCK* (or λ) and *VDJCM* genes. The solution to the problem of secreting antibody with the same specificity as that present on the cell surface as a membrane Ig is found in a **differential splicing** mechanism. The initial nuclear μ chain RNA transcript includes sequences coding for **hydrophobic transmembrane regions** which enable the IgM to sit in the membrane where it acts as the B-cell receptor (BCR), but if these are spliced out, the antibody molecules can be secreted in a soluble form (figure 4.1).

As the B-cell matures, it coexpresses a BCR utilizing surface IgD of the same specificity. This surface IgM+surface IgD B-cell phenotype is abundant in the mantle zone lymphocytes of secondary lymphoid follicles (cf. figure 8.7f) and is achieved by differential splicing of a single transcript containing VDJ, $C\mu$ and $C\delta$ segments producing either membrane IgM or IgD (figure 4.2). As the B-cell matures further, other isotypes such as IgG may be utilized in the BCR (cf. p. 238).

The surface immunoglobulin is complexed with associated membrane proteins

The cytoplasmic tail of the surface IgM is a miserable

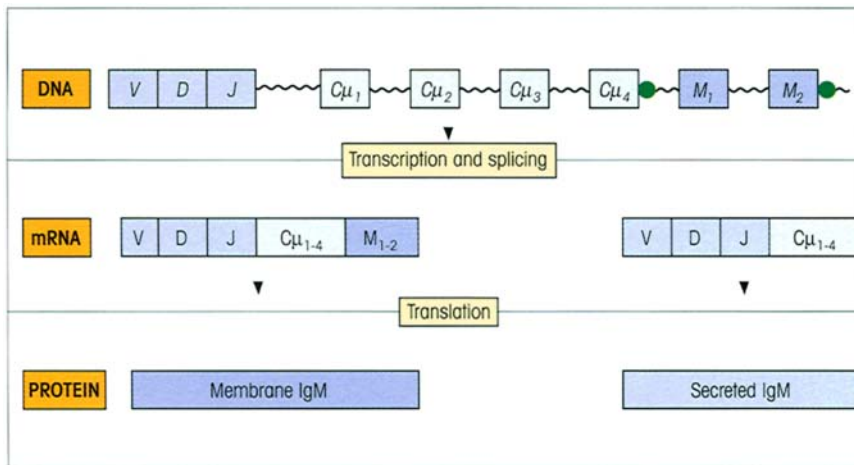


Figure 4.1. Splicing mechanism for the switch from the membrane to the secreted form of IgM. Alternative processing determines whether a secreted or membrane-bound form of the μ heavy chain is produced. If transcription termination or cleavage occurs in the intron between C μ_4 and M₁, the C μ_4 poly-A addition signal (AAUAAA) is used and the secreted form is produced. If transcription continues through the membrane exons, then C μ_4 can be spliced to the M sequences resulting in the M₂ poly-A addition signal being utilized. The hydrophobic sequence encoded by the exons M₁ and M₂ then anchors the receptor IgM to the membrane. For simplicity, the leader sequence has been omitted. ~ = introns.

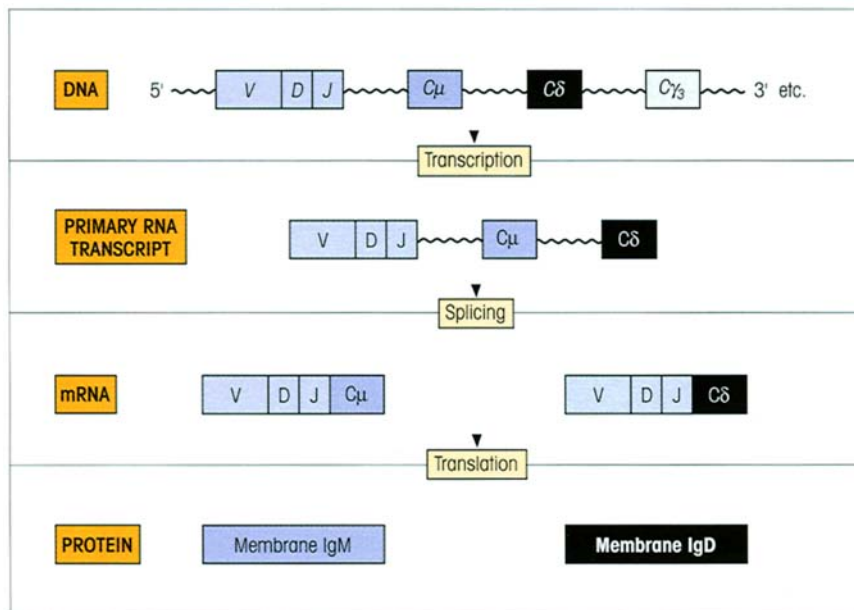


Figure 4.2. Surface membrane IgM and IgD receptors of identical specificity appear on the same cell through differential splicing of the composite primary RNA transcript (leader sequences again omitted for simplicity).

three amino acids long. In no way could this accommodate the structural motifs required for interaction with intracellular protein tyrosine kinases which mediate the activation of signal transduction cascades. With some difficulty, it should be said, it eventually proved possible to isolate a disulfide-linked heterodimer which copurifies with the membrane Ig. This consists of two glycoprotein chains called Ig- α (CD79a) and Ig- β (CD79b) (figure 4.3). Both Ig- α and Ig- β have an extracellular immunoglobulin-type domain, but it is their C-terminal cytoplasmic domains which are obligatory for signaling and which become phosphorylated on cell activation by antigen-induced cross-linking of the BCR, an event also associated with rapid Ca^{2+} mobilization. Tyrosine-containing structural motifs (immunoreceptor tyrosine-based activation motifs,

ITAMs) are present in the cytoplasmic domains of the Ig- α/β heterodimer (figure 4.3) and it is these that undergo phosphorylation by tyrosine kinases.

THE T-CELL SURFACE RECEPTOR FOR ANTIGEN (TCR)

The receptor for antigen is a transmembrane heterodimer

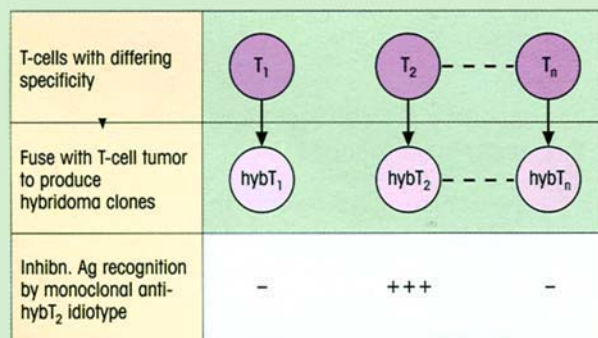
When it was eventually tracked down (Milestone 4.1), the antigen-specific **T-cell receptor** was identified as a membrane-bound molecule composed of two disulfide-linked chains, α and β . Each chain folds into two Ig-like domains, one having a relatively invariant structure and the other exhibiting a high degree of

Milestone 4.1—The T-cell Receptor

Since T-lymphocytes respond by activation and proliferation when they contact antigen presented by cells such as macrophages, it seemed reasonable to postulate that they do so by receptors on their surface. In any case, it would be difficult to fit T-cells into the clonal selection club if they lacked such receptors. Guided by Ockam's razor (the *Law of Parsimony*, which contends that it is the aim of science to present the facts of nature in the simplest and most economical conceptual formulations), most investigators plumped for the hypothesis that nature would not indulge in the extravagance of evolving two utterly separate molecular recognition species for B- and T-cells, and many fruitless years were spent looking for the Holy Grail of the T-cell receptor with anti-immunoglobulin serums or monoclonal antibodies (cf. p. 120). Success only came when a monoclonal antibody directed to the idiotype of a T-cell was used to block the response to antigen. This was identified by its ability to block one individual T-cell clone out of a large number, and it was correctly assumed

that the structure permitting this selectivity would be the combining site for antigen on the T-cell receptor. Immunoprecipitation with this antibody brought down a disulfide-linked heterodimer composed of 40–44 kDa subunits (figure M4.1.1).

The other approach went directly for the genes, arguing as follows. The T-cell receptor should be an integral membrane protein not present in B-cells. Hence, T-cell polyosomal mRNA from the endoplasmic reticulum, which should provide an abundant source of the appropriate transcript, was used to prepare cDNA from which genes common to B- and T-cells were subtracted by hybridization to B-cell mRNA. The resulting T-specific clones were used to probe for a T-cell gene which is rearranged in all functionally mature T-cells but is in its germ-line configuration in all other cell types (figure M4.1.2). In such a way were the genes encoding the β -subunit of the T-cell receptor uncovered.



Ab to T-cell receptor immunoprecipitates 2 chains

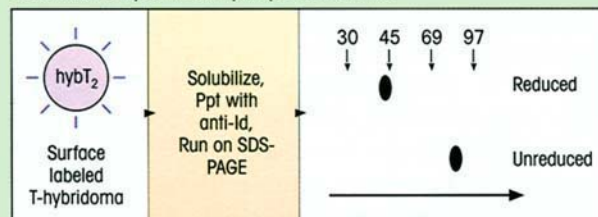


Figure M4.1.1. Ab to T-cell receptor (anti-idiotype) blocks Ag recognition. Based on Haskins K., Kubo R., White J., Pigeon M., Kappler J. & Marack P. (1983) *Journal of Experimental Medicine* 157, 1149. (Simplified a little.)

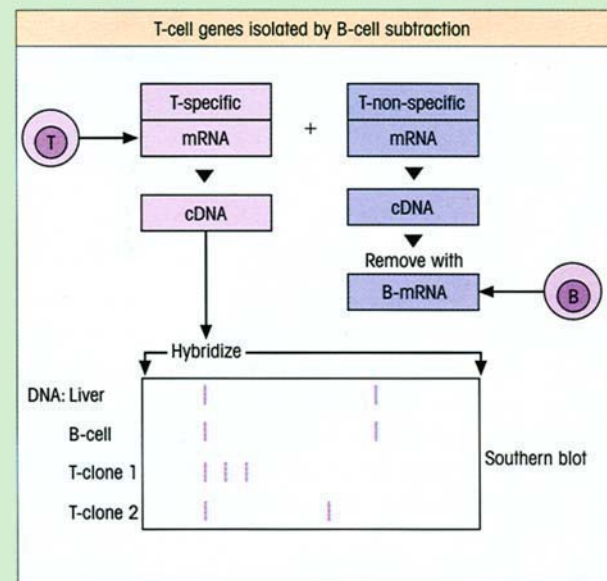


Figure M4.1.2. Isolation of T-cell receptor genes. Different sized DNA fragments produced by a restriction enzyme are separated by electrophoresis and probed with the T-cell gene. The T-cells show rearrangement of one of the two germ-line genes found in liver or B-cells. Based on Hendrick S.M., Cohen D.I., Nielsen E.A. & Davis M.M. (1984) *Nature* 308, 149.

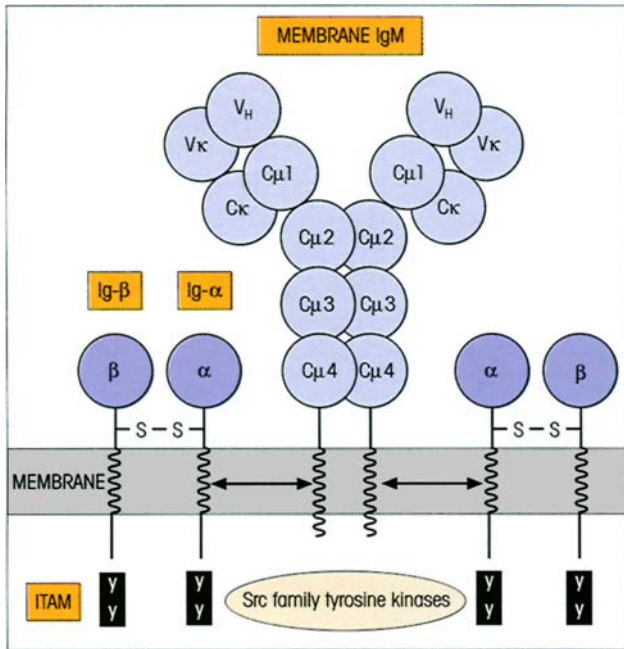


Figure 4.3. Model of B-cell receptor (BCR) complex. The Ig- α /Ig- β heterodimer is encoded by the B-cell-specific genes *mb-1* and *B29*, respectively. Two of these heterodimers are shown with the Ig- α associating with the membrane-spanning region of the IgM μ chain. The Ig-like extracellular domains are colored blue. Each tyrosine (Y)-containing box possesses a sequence of general structure Tyr.X₂.Leu.X₇.Tyr.X₂.Ile (where X is not a conserved residue), referred to as the immunoreceptor tyrosine-based activation motif (ITAM). On activation of the B-cell, these ITAM sequences act as signal transducers through their ability to associate with and be phosphorylated by a series of tyrosine kinases. Note that whilst a κ light chain is illustrated for the surface IgM, some B-cells utilize a λ light chain.

variability, so that the $\alpha\beta$ TCR has a structure really quite closely resembling an Ig Fab fragment. This analogy stretches even further—each of the two variable regions has three hypervariable regions which X-ray diffraction data have defined as incorporating the amino acids which make contact with the peptide-major histocompatibility complex (MHC) ligand.

Both α and β chains are required for antigen specificity as shown by transfection of the T-receptor genes from a cytotoxic T-cell clone specific for fluorescein to another clone of different specificity; when it expressed the new α and β genes, the transfected clone acquired the ability to lyse the fluoresceinated target cells. Another type of experiment utilized T-cell hybridomas formed by fusing single antigen-specific T-cells with T-cell tumors to achieve ‘immortality’. One hybridoma recognizing chicken ovalbumin, presented by a macrophage, gave rise spontaneously

to two variants, one of which lost the chromosome encoding the α chain, and the other, the β chain. Neither variant recognized antigen but, when they were physically fused together, each supplied the complementary receptor chain and reactivity with antigen was restored.

There are two classes of T-cell receptors

Not long after the breakthrough in identifying the $\alpha\beta$ T-cell receptor, came reports of the existence of a second type of receptor composed of γ and δ chains. Since it appears earlier in thymic ontogeny, the $\gamma\delta$ receptor is sometimes referred to as **TCR1** and the $\alpha\beta$ receptor as **TCR2** (cf. p. 227).

In the human, $\gamma\delta$ cells make up only 0.5–15% of the T-cells in peripheral blood but they show greater dominance in the intestinal epithelium and in skin. In contrast, between 30% and 80% of blood T-cells in ruminants are $\gamma\delta$, reflecting a somewhat different physiological life-style, but it does make the point that these cells can play an important role in immune responses. In general, $\gamma\delta$ T-cells seem to be strongly biased towards the recognition of certain types of microbial antigens, a number of which are components of mycobacteria, including lipid and glycolipid molecules and heat-shock proteins.

The encoding of T-cell receptors is similar to that of immunoglobulins

The gene segments encoding the T-cell receptor β chains follow a broadly similar arrangement of *V*, *D*, *J* and constant segments to that described for the immunoglobulins (figure 4.4). In a parallel fashion, as an immunocompetent T-cell is formed, rearrangement of *V*, *D* and *J* genes occurs to form a continuous *VDJ* sequence. The firmest evidence that B- and T-cells use similar recombination mechanisms comes from mice with severe combined immunodeficiency (SCID) which have a single autosomal recessive defect preventing successful recombination of *V*, *D* and *J* segments (cf. p. 42). Homozygous mutants fail to develop immunocompetent B- and T-cells and identical sequence defects in *VDJ* joint formation are seen in both pre-B- and pre-T-cell lines.

Looking first at the β chain cluster, one of the two *D β* genes rearranges next to one of the *J β* genes. Note that, because of the way the genes are organized, the first *D β* gene, *D β ₁*, can utilize any of the 13 *J β* genes, but *D β ₂* can only choose from the seven *J β ₂* genes (figure 4.4). Next one of the 50 or so *V β* genes is rearranged to the pre-

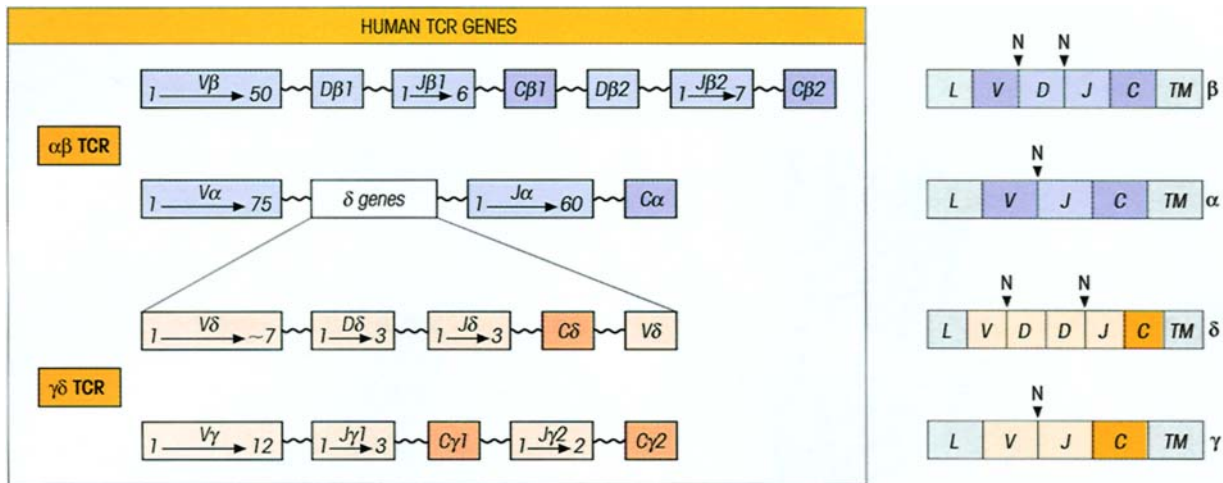


Figure 4.4. Genes encoding $\alpha\beta$ and $\gamma\delta$ T-cell receptors. Genes encoding the δ chains lie between the $V\alpha$ and $J\alpha$ clusters and some V segments in this region can be used in either δ or α chains, i.e. as either $V\alpha$ or $V\delta$. TCR genes rearrange in a manner analogous to that seen with immunoglobulin genes, including N-region diversity at the $V(D)J$ junctions (cf. figures 3.8 and 3.10). One of the $V\delta$ genes is found downstream (3') of the $C\delta$ gene and rearranges by an inversionsal mechanism.

formed $D\beta J\beta$ segment. **Variability in junction formation** and the **random insertion of nucleotides** to create N-region diversity either side of the D segment mirror the same phenomenon seen with Ig gene rearrangements. Sequence analysis emphasizes the analogy with the antibody molecule; each V segment contains two hypervariable regions, while the DJ junctional sequence provides the **very hypervariable** CDR3 structure, making a total of six potential complementarity determining regions for antigen binding in each TCR (figure 4.5). As in the synthesis of antibody, the intron between VDJ and C is spliced out of the mRNA before translation with the restriction that rearrangements involving genes in the $D\beta J\beta$ cluster can only link to $C\beta_2$.

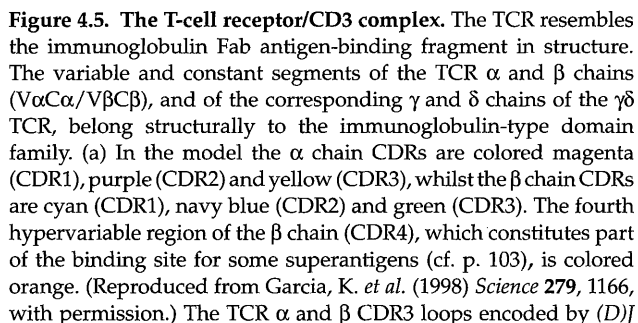
All the other chains of the TCRs are encoded by genes formed through similar translocations. The α chain gene pool lacks D segments but possesses a prodigious number of J segments. The number of $V\gamma$ and $V\delta$ genes is small in comparison with $V\alpha$ and $V\beta$. Like the α chain pool, the γ chain cluster has no D segments. The awkward location of the δ locus embedded within the α gene cluster results in T-cells which have undergone $V\alpha$ - $J\alpha$ combination having no δ genes on the rearranged chromosome; in other words, the δ genes are completely excised.

The CD3 complex is an integral part of the T-cell receptor

The T-cell antigen recognition complex and its B-cell counterpart can be likened to specialized army platoons whose job is to send out a signal when the enemy has been sighted. When the TCR 'sights the enemy', i.e. ligates antigen, it relays a signal through an associated complex of transmembrane polypeptides (**CD3**) to the interior of the T-lymphocyte, instructing it to awaken from its slumbering G0 state and do something useful—like becoming an effector cell. In all immunocompetent T-cells, the antigen receptor is noncovalently but still intimately linked with CD3 in a complex which, as current wisdom has it, may contain two heterodimeric TCR $\alpha\beta$ or $\gamma\delta$ recognition units closely apposed to one molecule of the invariant CD3 polypeptide chains γ and δ , two molecules of CD3 ϵ , plus the disulfide-linked ζ - ζ dimer. The total complex therefore has the structure $\text{TCR}_2\text{-CD3}\gamma\delta\epsilon_2\text{-}\zeta_2$ (figure 4.5b). The ITAM tyrosine motifs (cf. legend figure 4.3) associate with protein tyrosine kinases thereby transducing signals generated by ligand binding to the TCR. In mice, either or both of the ζ chains can be replaced by a splice variant from the ζ gene termed η . The ζ chain also associates with the $\text{Fc}\gamma\text{RIIIA}$ receptor in natural killer (NK) cells where it functions as part of the signal transduction mechanism in that context also.

THE GENERATION OF DIVERSITY FOR ANTIGEN RECOGNITION

We know that the immune system has to be capable of recognizing virtually any pathogen that has arisen or might arise. The awesome genetic solution to this problem of anticipating an unpredictable future



genes are both short; the TCR γ CDR3 is also short with a narrow length distribution, but the δ loop is long with a broad length distribution, resembling the Ig light and heavy chain CDRs, respectively. (b) The TCRs may be expressed in pairs linked to the CD3 complex. Negative charges on transmembrane segments of the invariant chains of the CD3 complex contact the opposite charges on the TCR $C\alpha$ and $C\beta$ chains conceivably as depicted. (c) The cytoplasmic domains of the CD3 peptide chains contain immunoreceptor tyrosine-based activation motifs (ITAM; cf. B-cell receptor, figure 4.3) which contact src protein tyrosine kinases. Try not to confuse the TCR $\gamma\delta$ and the CD3 $\gamma\delta$ chains.

Intrachain amplification of diversity

Random VDJ combination increases diversity geometrically

Just as we can use a relatively small number of different building units in a child's construction set such as Lego to create a rich variety of architectural masterpieces, so the individual receptor gene segments can be viewed as building blocks to fashion a multiplicity of antigen-specific receptors for both B- and T-cells.

The light chain variable regions are created from *V* and *J* segments, and the heavy chain variable regions from *V*, *D* and *J* segments. As already described in the previous chapter, the enzymes RAG-1 and RAG-2 recognize recombination signal sequences (RSSs) adjacent to the coding sequences of these gene segments. The RSSs consist of conserved heptamers and nonamers separated by spacers of either 12 or 23 base pairs (cf. p. 43) and are found at the 3' of each *V* segment, on both the 5' and 3' sides of each *D* segment, and at the 5' of each *J* segment (see figure 4.8). Those associated with the *V_H* and *J_H* segments have 23 base pair spacers; those flanking the *D_H* segments have 12 base pair spacers. An RSS with a 23 base pair spacer can only recombine with an RSS containing a 12 base pair spacer, the so-called '12/23 rule'. Thus the arrangement of the RSSs of the immunoglobulin heavy chain genes ensures that a *D* segment is always included in the rearrangement; *V_H* cannot join directly to *J_H* as they both have 23 base pair spacers. A comparable enforcement exists for the *V*, *D* and *J* segments of the TCR β chain.

To see how sequence diversity is generated, let us take the immunoglobulin heavy chain genes as an example (table 4.1). Although the precise number of gene segments varies from one individual to another, there are typically around 25 *D* and six *J* functional segments. If there were entirely **random joining** of any one *D* to any one *J* segment (cf. figure 3.8), we would in this individual have the possibility of generating 150 *DJ* combinations (25×6). Let us go to the next

stage. Since each of these 150 *DJ* segments could join with any one of the approximately 50 *V_H* functional sequences, the net potential repertoire of *VDJ* genes encoding the heavy chain variable region would be $50 \times 150 = 7500$. In other words, just taking the *V*, *D* and *J* genes, which in this example add up arithmetically to 81, we have produced a range of some 7500 different variable regions by **geometric recombination** of the basic elements. But that is only the beginning.

Playing with the junctions

Another ploy to squeeze more variation out of the germ-line repertoire involves variable boundary recombinations of *V*, *D* and *J* to produce different junctional sequences (figure 4.6).

As discussed earlier, further diversity results from the generation of palindromic sequences (P-elements, figure 4.7a) arising from the formation of hairpin structures during the recombination process and from the insertion of nucleotides at the N region between the *V*, *D* and *J* segments, a process associated with the expression of terminal deoxynucleotidyl transferase (figure 4.7b). Whilst these mechanisms add nucleotides to the sequence, yet more diversity can be created by nucleases chewing away at the exposed strand ends to remove nucleotides. These maneuvers greatly increase the repertoire, especially important for the T-cell receptor γ and δ genes which are otherwise rather limited in number.

Table 4.1. Calculations of human *V* gene diversity. It is known that the precise number of gene segments varies from one individual to another, perhaps between 40 and 70 in the case of the *V_H* genes for example, so that these calculations represent 'typical' numbers. The number of specificities generated by straightforward random combination of germ-line segments is calculated. These will be increased by the further mechanisms listed: *minimal assumption of approximately 10 variants for chains lacking *D* segments and 100 for chains with *D* segments. The calculation for the T-cell receptor β chain requires further explanation. The first of the two *D* segments, *D β_1* , can combine with 50 *V* genes and with all 13 *J β_1* and *J β_2* genes. *D β_2* behaves similarly but can only combine with the seven downstream *J β_2* genes.

	$\gamma\delta$ TCR (TCR1)		$\alpha\beta$ TCR (TCR2)		Ig		
	γ	δ	α	β	H	κ	L λ
<i>V</i> gene segments	12	~8	75	50	50	40	30
<i>D</i> gene segments	-	3	-	1,1	25	-	-
<i>J</i> gene segments	3,2	3	60	6,7	6	5	4
Random Combinatorial joining (without junctional diversity)	<i>V</i> × <i>J</i>	<i>V</i> × <i>D</i> × <i>J</i>	<i>V</i> × <i>J</i>	<i>V</i> × <i>D</i> × <i>J</i>	<i>V</i> × <i>D</i> × <i>J</i>	<i>V</i> × <i>J</i>	<i>V</i> × <i>J</i>
	12 × 5	8 × 3 × 3	75 × 60	50(13+7)	50 × 25 × 6	40 × 5	30 × 4
Total	60	72	4500	1000	7500	200	120
Combinatorial heterodimers	60 × 72		4500 × 1000		7500 × 200		7500 × 120
Total (rounded)	4.3×10^3		4.5×10^6		1.5×10^6		0.9×10^6
Other mechanisms: D's in 3 reading frames, junctional diversity, N region insertion; * × 10^3	4.3×10^6		4.5×10^9		1.5×10^9		0.9×10^9
Somatic mutation	-		-		+++		+++

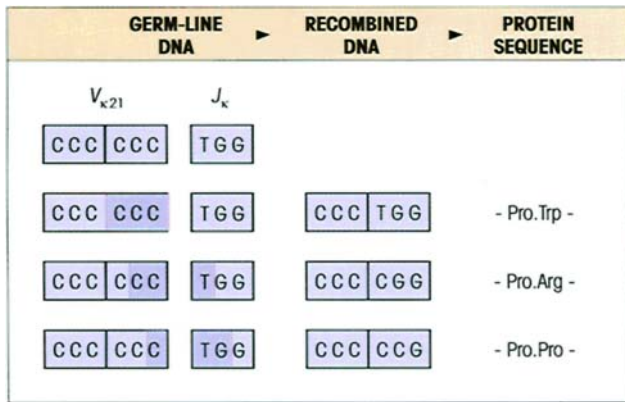


Figure 4.6. Junctional diversity between two germ-line segments producing three variant protein sequences. The nucleotide triplet which is spliced out is colored the darker blue.

Additional mechanisms relate specifically to the *D*-region sequence: particularly in the case of the TCR δ genes, where the *D* segment can be read in three different reading frames and two *D* segments can join together, such *DD* combinations produce a longer third complementarity determining region (CDR3) than is found in other TCR or antibody molecules.

Since the CDR3 in the various receptor chains is essentially composed of the regions between the *V(D)J* segments, where junctional diversity mechanisms can introduce a very high degree of amino acid variability, one can see why it is that this hypervariable loop usually contributes the most to determining the fine antigen-binding specificity of these molecules.

Receptor editing

Recent observations have established that lymphocytes are not necessarily stuck with the antigen receptor they initially make; if they don't like it they can change it. The replacement of an undesired receptor with one which has more acceptable characteristics is referred to as receptor editing. This process has been described for both immunoglobulins and for TCR, allowing the replacement of either nonfunctional rearrangements or autoreactive specificities. Furthermore, receptor editing in the periphery may rescue low affinity B-cells from apoptotic cell death by replacing a low affinity receptor with a selectable one of higher affinity. That this does indeed occur in the periphery is strongly supported by the finding that mature B-cells in germinal centers can express RAG-1 and RAG-2 which mediate the rearrangement process.

But how does this receptor editing work? Well, in the case of the receptor chains which lack *D* gene seg-

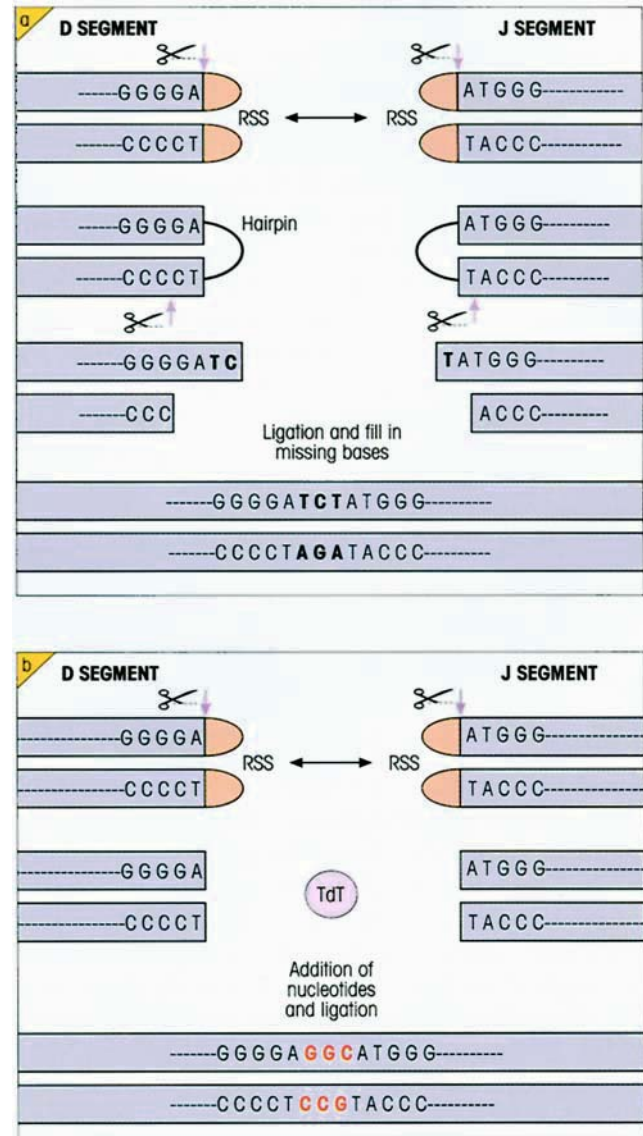


Figure 4.7. P-elements and N-region diversity. In addition to junctional diversity (cf. figure 4.6), additional nucleotides can be incorporated into the final sequence. (a) This can occur when hairpin loop structures are created between the two strands of the DNA following cleavage at the RSS, and a subsequent cleavage of one strand creates an overhang which acts as a template for the addition of nucleotides, creating a Palindromic sequence (P-element, here GATC/CTAG and TA/AT, with the newly added sequence in bold). (b) Nucleotides can be added in a Nontemplated fashion (N-region diversity, indicated by the red nucleotides) by the enzyme terminal deoxynucleotidyl transferase (TdT).

ments, namely the immunoglobulin light chain and the TCR α chain, a secondary rearrangement may occur by a *V* gene segment upstream of the previously rearranged *VJ* segment recombining to a 3' *J* gene sequence, both of these segments having intact RSSs that are compatible (figure 4.8a). However, for immunoglobulin heavy chains and TCR β chains the process of *VDJ* rearrangement deletes all of the *D*

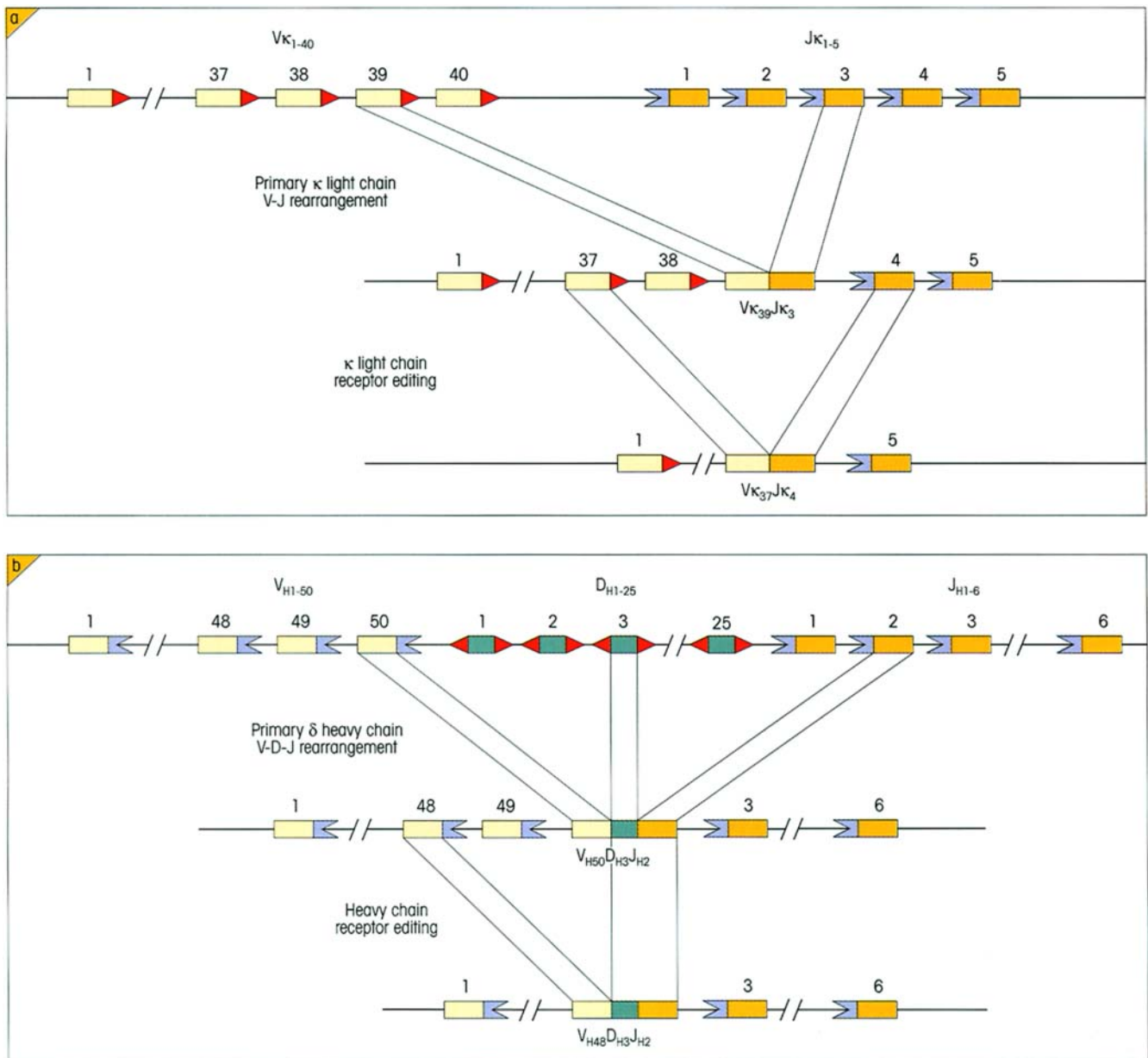


Figure 4.8. Receptor editing. (a) For immunoglobulin light chain or TCR α chain the recombination signal sequences (RSSs; heptamer–nonamer motifs) at the 3′ of each variable (V) segment and the 5′ of each joining (J) segment are compatible with each other and therefore an entirely new rearrangement can potentially occur as shown. This would result in a receptor with a different light chain variable sequence (in this example V κ_{37} /J κ_4 replacing V κ_{39} /J κ_3) together with the original heavy chain. (b) With respect to the immunoglobulin heavy chain or TCR β chain the organization of the heptamer–nonamer sequences in the RSS precludes a V segment directly recombining with the J segment. This is the so-called 12/23 rule whereby the

heptamer–nonamer sequences associated with a 23 base pair spacer (colored violet) can only base pair with heptamer–nonamer sequences containing a 12 base pair spacer (colored red). The heavy chain V and J both have an RSS with a 23 base pair spacer and so this is a nonstarter. Furthermore, all the unrearranged D segments have been deleted so that there are no 12 base pair spacers remaining. This apparent bar to secondary rearrangement is probably overcome by the presence of an RSS-like sequence near the 3′ end of the V gene coding sequences, so that only the V gene segment is replaced (in the example shown, the sequence V H_{48} D H_{3} J H_2 replaces V H_{50} D H_{3} J H_2).

segment-associated RSSs (figure 4.8b). Because V H and J H both have 23 base pair spacers in their RSSs, they cannot recombine: that would break the 12/23 rule. This apparent obstacle to receptor editing of these chains may be overcome by the presence of a sequence

near the 3′ end of the V coding sequences that can function as a surrogate RSS, such that the new V segment would simply replace the previously rearranged V, maintaining the same D and J sequence (figure 4.8b). This is probably a relatively inefficient process and

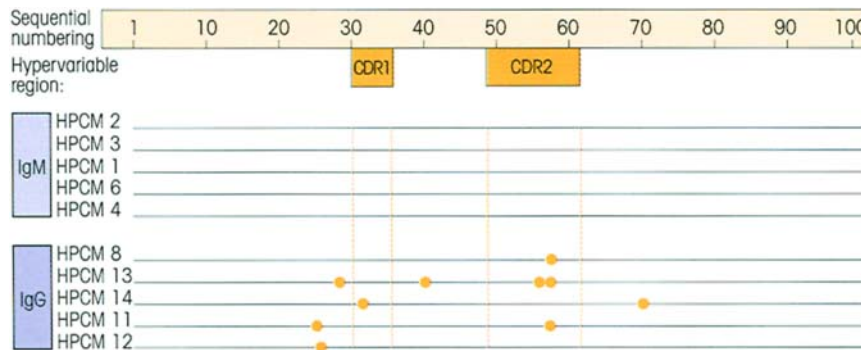


Figure 4.9. Mutations in a germ-line gene. The amino acid sequences of the V_H regions of five IgM and five IgG monoclonal phosphorylcholine antibodies generated during an antipneumococcal response in a single mouse are compared with the primary structure of the T15 germ-line sequence. A line indicates identity with the T15 prototype and an orange circle a single amino acid difference. Mutations have only occurred in the IgG molecules and are seen in both hypervariable and framework segments. (After Gearhart P.J. (1982) *Immunology Today* 3, 107.) Whilst in some other studies somatic hypermutation has been seen in IgM antibodies, the amount of mutation usually greatly increases following class switching.

receptor editing may therefore occur more readily in immunoglobulin light chains and TCR α chains than in immunoglobulin heavy chains and TCR β chains. Indeed, it has been suggested that the TCR α chain may undergo a series of rearrangements, continuously deleting previously functionally rearranged VJ segments until a selectable TCR is produced.

Interchain amplification

The immune system took an ingenious step forward when two different types of chain were utilized for the recognition molecules because the combination produces not only a larger combining site with potentially greater affinity, but also new variability. Heavy–light chain pairing amongst immunoglobulins appears to be largely random and therefore two B-cells can employ the same heavy chain but different light chains. This route to producing antibodies of differing specificity is easily seen *in vitro* where shuffling different recombinant light chains against the same heavy chain can be used to either fine tune or sometimes even alter the specificity of the final antibody. In general, the available evidence suggests that *in vivo* the major contribution to diversity and specificity comes from the heavy chain, perhaps not unrelated to the fact that the heavy chain CDR3 gets off to a head start in the race for

diversity being, as it is, encoded by the junctions between three gene segments: V , D and J .

This random association between TCR γ and δ chains, TCR α and β chains, and Ig heavy and light chains yields a further geometric increase in diversity. From table 4.1 it can be seen that approximately 230 functional T-cell receptor and 160 functional Ig germ-line segments can give rise to 4.5 million and 2.4 million different combinations, respectively, by straightforward associations *without* taking into account all of the fancy junctional mechanisms described above. Hats off to evolution!

Somatic hypermutation

There is inescapable evidence that immunoglobulin V -region genes can undergo significant **somatic hypermutation**. Analysis of 18 murine λ myelomas revealed 12 with identical structure, four showing just one amino acid change, one with two changes and one with four changes, all within the hypervariable regions and indicative of somatic hypermutation of the single mouse λ germ-line gene. In another study, following immunization with pneumococcal antigen, a single germ-line T15 V_H gene gave rise by mutation to several different V_H genes all encoding phosphorylcholine antibodies (figure 4.9).

A number of features of this somatic diversification phenomenon deserve a mention. The mutations are the result of single nucleotide substitutions, they are restricted to the variable as distinct from the constant region and occur in both framework and hypervariable regions. The mutation rate is remarkably high, approximately 10^{-4} – 10^{-3} per base pair per generation, which is approximately a million times higher than for other mammalian genes. In addition, the mutational mechanism is bound up in some way with class switch since hypermutation is more frequent in IgG and IgA than in IgM antibodies, affecting both heavy (figure

4.9) and light chains. However, V_H genes are on average more mutated than V_L genes. This might be a consequence of receptor editing acting more frequently on light chains, as this would have the effect of wiping the slate clean with respect to light chain V gene mutations whilst maintaining already accumulated heavy chain V gene point mutations.

Somatic hypermutation does not appear to add significantly to the repertoire available in the early phases of the primary response, but occurs during the generation of memory and is probably responsible for tuning the response towards higher affinity. The mechanism behind the greatly enhanced mutation frequency in immunoglobulin genes is unknown, but, intriguingly, it has recently been shown that the *bcl-6* proto-oncogene also undergoes somatic hypermutation in around one-third of normal germinal center B-cells and memory B-cells. The rate of mutation is only around 10% of that seen in the immunoglobulin genes, but nonetheless is way above that seen in other genes in the B-cell. Furthermore, the types of mutation are extremely similar to those seen in the immunoglobulin V genes, suggesting a common mechanism. The present favored idea is that somatic hypermutation is associated with an error-prone DNA polymerase which is coupled to RNA transcription.

Recently data have been put forward suggesting that there is yet another mechanism for creating further diversity. This involves the insertion or deletion of short stretches of nucleotides within the immunoglobulin V gene sequence of both heavy and light chains. This mechanism would have an intermediate effect on antigen recognition, being more dramatic than single point mutation, but considerably more subtle than receptor editing. In one study, a reverse transcriptase-polymerase chain reaction (RT-PCR) was employed to amplify the expressed V_H and V_L genes from 365 IgG⁺ B-cells and it was shown that 6.5% of the cells contained nucleotide insertions or deletions. The transcripts were left in-frame and no stop codons were introduced by these modifications. The percentage of cells containing these alterations is likely to be an underestimate. All the insertions and deletions were in, or near to, CDR1 and/or CDR2. N-region diversity of the CDR3 meant that it was not possible to analyse the third hypervariable region for insertions/deletions of this type and therefore these would be missed in the analysis. The fact that the alterations were associated with CDRs does suggest that the B-cells had been subjected to selection by antigen. It was also notable that the insertions/deletions occurred at known hot spots for somatic point mutation, and the same error-prone DNA polymerase responsible for somatic hypermuta-

tion may also be involved here. The sequences were often a duplication of an adjacent sequence in the case of insertions or a deletion of a known repeated sequence. This type of modification may, like receptor editing, play a major role in eliminating autoreactivity and also in enhancing antibody affinity.

T-cell receptor genes, on the other hand, **do not generally undergo somatic hypermutation**. It has been argued that this would be a useful safety measure since T-cells are positively selected in the thymus for weak reactions with self MHC (cf. p. 226), so that mutations could readily lead to the emergence of high affinity autoreactive receptors and autoimmunity.

One may ask how it is that this array of germ-line genes is protected from genetic drift. With a library of 390 or so functional V , D and J genes, selection would act only weakly on any single gene which had been functionally crippled by mutation and this implies that a major part of the library could be lost before evolutionary forces operated. One idea is that each sub-family of related V genes contains a prototype coding for an antibody indispensable for protection against some common pathogen, so that mutation in this gene would put the host at a disadvantage and would therefore be selected against. If any of the other closely related genes in its set became defective through mutation, this indispensable gene could repair them by gene conversion, a mechanism in which it will be remembered that two genes interact in such a way that the nucleotide sequence of part or all of one becomes identical to that of the other. Although gene conversion has been invoked to account for the diversification of MHC genes, it can also act on other families of genes to maintain a degree of sequence homogeneity. Certainly it is used extensively by, for example, chickens and rabbits, in order to generate immunoglobulin diversity. In the rabbit only a single germ-line V_H gene is rearranged in the majority of B-cells; this then becomes a substrate for gene conversion by one of the large number of V_H pseudogenes. There are also large numbers of V_H pseudogenes and orphan genes (genes located outside the gene locus, often on a completely different chromosome) in humans which actually outnumber the functional genes, although there is no evidence to date that these are used in gene conversion processes.

NK RECEPTORS

One of the main functions of natural killer (NK) cells is to patrol the body looking for cells which have lost expression of the normally ubiquitously present MHC class I molecules. Such abnormal cells are usually either malignant or infected with a microorganism that

interferes with class I expression. The NK cells carry out their task using two sets of receptors, *activating receptors* that recognize molecules collectively present on all cell surfaces and *inhibitory receptors* that recognize MHC class I molecules. Upon ligation, the activating receptors signal the NK cell to kill the target cell and/or to secrete cytokines. This potentially anarchic situation in which the NK cells would attack all cells in the body is normally prevented due to the recognition of MHC class I by the inhibitory receptors (figure 4.10). Any nucleated cell lacking MHC class I will not engage the inhibitory receptors and will only trigger the activating receptors, resulting in its execution by the NK cell.

A second mode of killing which NK cells enjoy is antibody-dependent cellular cytotoxicity (ADCC, cf. p. 32) for which they are equipped with Fc γ RIII receptors in order to recognize antibody-coated target cells (figure 4.10).

THE MAJOR HISTOCOMPATIBILITY COMPLEX (MHC)

Molecules within this complex were originally defined by their ability to provoke vigorous rejection of grafts exchanged between different members of a species (Milestone 4.2). In Chapter 2, brief mention was made of the necessity for antigens to be associated with class I or class II MHC molecules in order that they may be recognized by T-lymphocytes. The intention now is to give more insight into the nature of these molecules.

Class I and class II molecules are membrane-bound heterodimers

MHC class I

Class I molecules consist of a heavy polypeptide chain of 44 kDa noncovalently linked to a smaller 12 kDa polypeptide called β_2 -microglobulin. The largest part of the heavy chain is organized into three globular domains (α_1 , α_2 and α_3 ; figure 4.11) which protrude from the cell surface; a hydrophobic section anchors the molecule in the membrane and a short hydrophilic sequence carries the C-terminus into the cytoplasm.

X-ray analysis of crystals of a human class I molecule provided an exciting leap forwards in our understanding of MHC function. Both β_2 -microglobulin and the α_3 region resemble classic Ig domains in their folding pattern (cf. figure 4.11c). However, the α_1 and α_2 domains, which are most distal to the membrane, form two

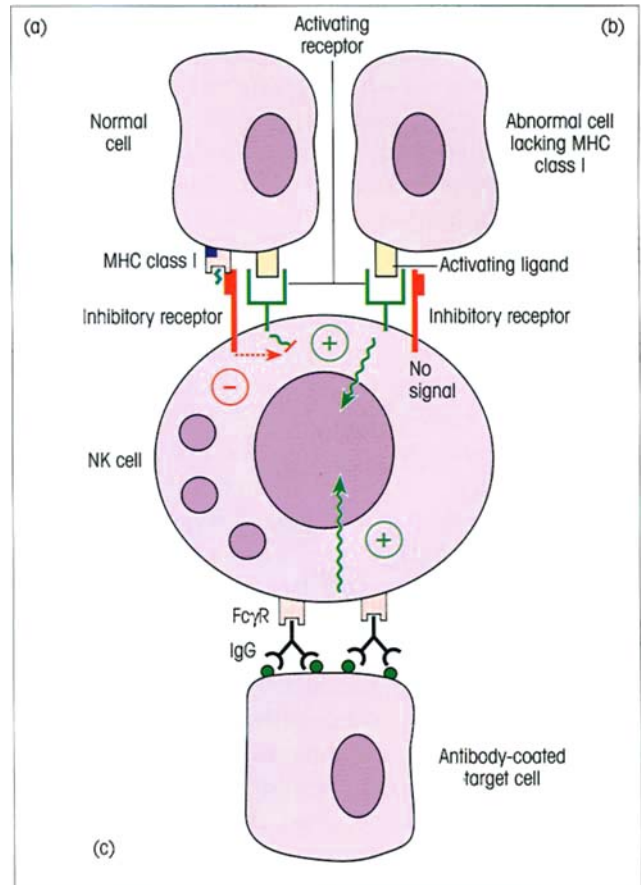


Figure 4.10. Natural killer (NK) cells. NK cells possess both activating and inhibitory receptors. Engagement of an activating receptor by its ligand sends a stimulatory signal into the NK cell but this is normally subverted by a signal transmitted upon recognition of MHC class I molecules by an inhibitory receptor (a). Any nucleated cell lacking class I is deemed abnormal and is killed following unimpeded transmission of the activation signal (b). Many NK receptors belong to either the killer cell immunoglobulin-like receptor (KIR) family or the C-type lectin domain family. Both families contain some members that are activating and others that are inhibitory. The inhibitory receptors bear immunoreceptor tyrosine-based inhibition motifs (ITIMs) in their cytoplasmic domain. Activating receptors lack ITIMs and possess a charged residue in their transmembrane sequence which permits their association with immunoreceptor tyrosine-based activation motif (ITAM)-containing adaptor proteins such as DAP12, CD3 ζ chain or Fc γ -chain. With respect to the KIR family, isoforms with a long cytoplasmic domain (KIR2DL and KIR3DL, depending on whether they possess 2 or 3 Ig-type extracellular domains) bear ITIMs and are therefore inhibitory, whilst those with a short cytoplasmic domain (KIR2DS and KIR3DS) lack ITIMs, associate with DAP12, and are therefore activating. Inhibitory KIRs recognize HLA-A, -B and -C irrespective of the peptide bound, whilst some of the inhibitory C-type lectin NK receptors interact with HLA-E presenting signal sequence peptides derived from other HLA class I molecules. C-type lectin family members which are inhibitory include CD94/NKG2A and, in the mouse, NKR-P1B and Ly-49C, whilst those that are activating include CD94/NKG2C, CD94/NKG2D, and the murine Ly-49D. The stress-inducible nonclassical MHC molecule MICA is a major ligand for NKG2D. NK cells usually possess several different inhibitory and activating receptors and it is the balance of signals from these that determines whether the cell becomes activated. Like several other cell types, NK cells can utilize their Fc γ receptors to mediate ADCC on target cells coated with antibody (c).

Milestone 4.2 — The Major Histocompatibility Complex

Peter Gorer raised rabbit antisera to erythrocytes from pure strain mice (resulting from >20 brother–sister matings) and, by careful cross-absorption with red cells from different strains, he identified the strain-specific antigen II, now known as H-2 (table M4.2.1).

He next showed that the rejection of an albino (A) tumor by black (C57) mice was closely linked to the presence of the antigen II (table M4.2.2) and that tumor rejection was associated with the development of antibodies to this antigen.

Subsequently, George Snell introduced the term **histocompatibility (H)** antigen to describe antigens provoking graft rejection and demonstrated that, of all the potential H antigens, differences at the H-2 (i.e. antigen II) locus provoked the strongest graft rejection seen between various mouse strains. *Poco a poco*, the painstaking studies gradually uncovered a remarkably complicated situation. Far

from representing a single gene locus, H-2 proved to be a large complex of multiple genes, many of which were highly polymorphic, hence the term **major histocompatibility complex (MHC)**. The major components of the current genetic maps of the human HLA and mouse H-2 MHC are drawn in figure M4.2.1 to give the reader an overall grasp of the complex make-up of this important region (to immunologists we mean!—presumably all highly transcribed regions are important to the host in some way).

Table M4.2.1. Identification of H-2 (antigen II).

Rabbit antiserum to:	Antigens detected on Albino red cells		
	I	II	III
Albino (A)	+++	+++	++
Black (C57)	++	–	++

Table M4.2.2. Relationship of antigen II to tumor rejection.

Antigen II phenotype of recipient strain	Rejection of tumor inoculum (A strain) by:			
	*Pure strain		**(A × C57) F1 backcross to C57	
	–	+	–	+
Ag II +ve (A)	39	0	17 (19.3)	17 (19.5)
Ag II -ve (C57)	0	45	0	44 (39)

*A tumor inoculum derived from A strain bearing antigen II is rejected by the C57 host (+ = rejection; – = acceptance).

**Offspring of A × C57 mating were backcrossed to the C57 parent and the resulting progeny tested for antigen II (Ag II) and their ability to reject the tumor. The figures in brackets = number expected if tumor growth is influenced by two dominant genes, one of which determines the presence of antigen II.

MAIN GENETIC REGIONS OF THE MAJOR HISTOCOMPATIBILITY COMPLEX												
HUMAN	MHC CLASS	II			III				I			CHROMOSOME 6
	HLA	DP	DQ	DR	C'	HSP	TNF	etc	B	C	A	
MOUSE	MHC CLASS	I	II		III				I		CHROMOSOME 17	
	H-2	K	A	E	C'	HSP	TNF	etc	D	L		

Figure M4.2.1. Main genetic regions of the major histocompatibility complex.

extended α -helices above a floor created by strands held together in a β -pleated sheet, the whole forming an undeniable **groove** (figure 4.11b and c). The appearance of these domains is so striking, we doubt whether the reader needs the help of gastronomic analogies such as ‘two sausages on a barbecue’ to prevent any

class I structural amnesia. Another curious feature emerged. The groove was occupied by a linear molecule, now known to be a peptide, which had cocrystallized with the class I protein. The significance of these unique findings for T-cell recognition of antigen will be revealed in the following chapter.

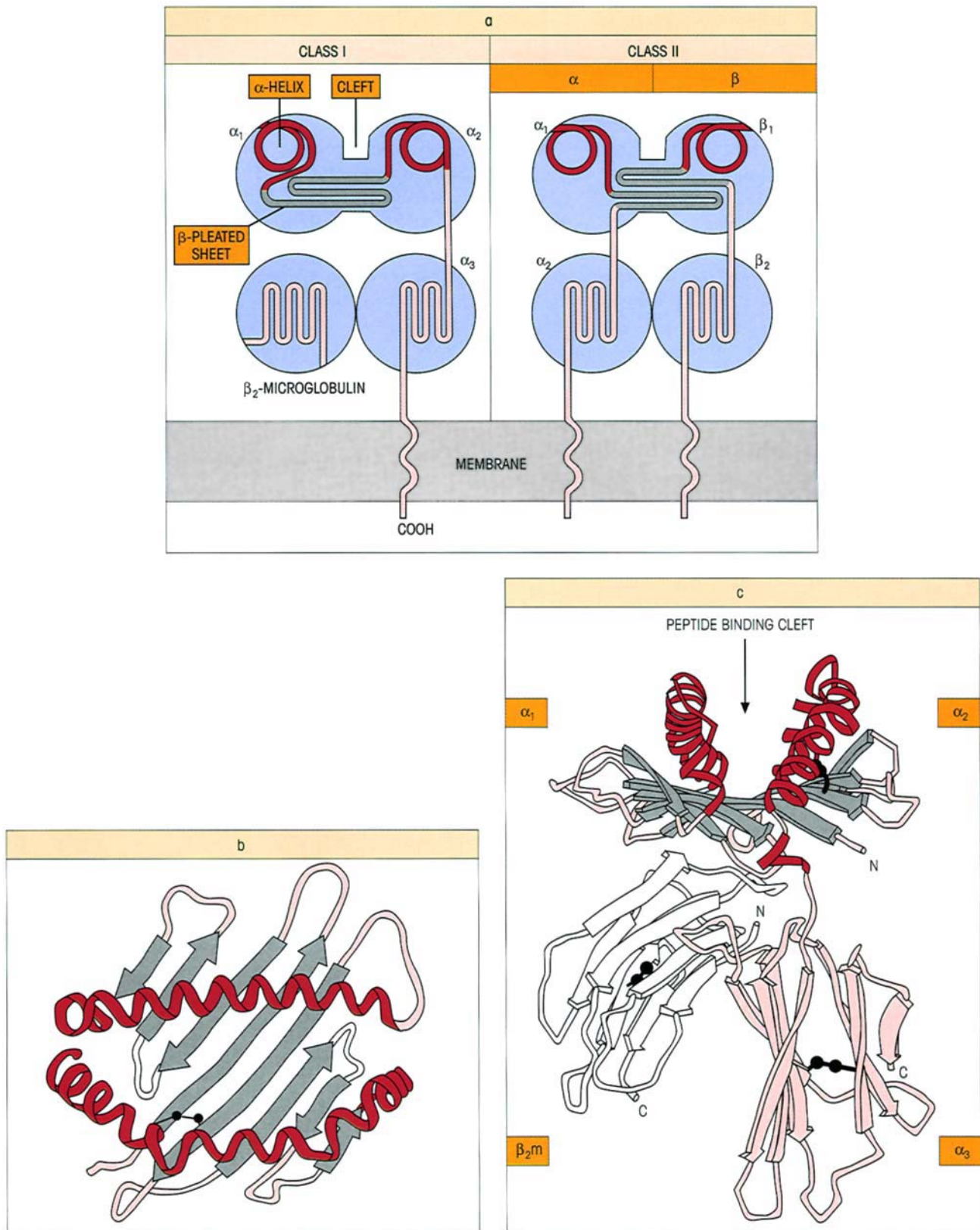


Figure 4.11. Class I and class II MHC molecules. (a) Diagram showing domains and transmembrane segments; the α -helices and β -sheets are viewed end on. (b) Schematic bird's eye representation of the top surface of human class I molecule (HLA-A2) based on the X-ray crystallographic structure. The strands making the β -pleated sheet are shown as thick gray arrows in the amino to carboxy direction; α -helices are represented as dark red helical ribbons. The

inside-facing surfaces of the two helices and the upper surface of the β -sheet form a cleft. The two black spheres represent an intrachain disulfide bond. (c) Side view of the same molecule clearly showing the anatomy of the cleft and the typical Ig-type folding of the α_3 - and β_2 -microglobulin domains (four antiparallel β -strands on one face and three on the other). (Reproduced from Bjorkman P.J. *et al.* (1987) *Nature* 329, 506, with permission.)

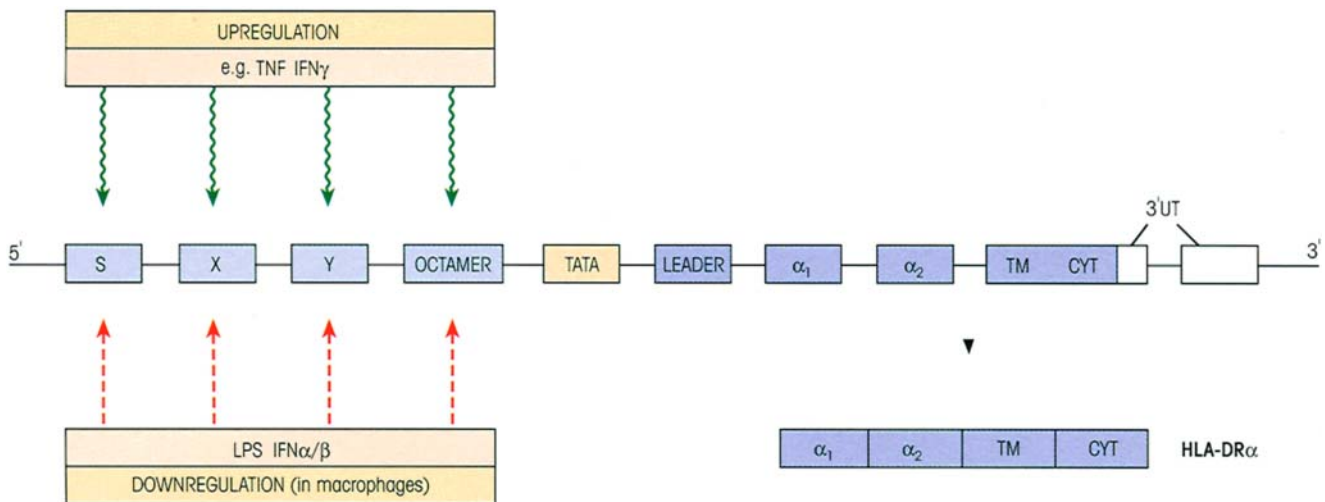


Figure 4.12. Genes encoding human HLA-DR α chain (darker blue) and their controlling elements (regulatory sequences in light blue and TATA box promoter in yellow). α_1/α_2 encode the two extracellular domains; TM and CYT encode the transmembrane and cytoplasmic segments, respectively. 3' UT represents the 3' untranslated sequence. Octamer motifs are also found in virtually all heavy and light chain immunoglobulin V gene promoters (cf. figure 3.6) and in the promoters of other B-cell-specific genes such as *B29* and *CD20*.

MHC class II

Class II MHC molecules are also transmembrane glycoproteins, in this case consisting of α and β polypeptide chains of molecular weight 34 kDa and 29 kDa, respectively.

There is considerable sequence homology with class I and structural studies have shown that the α_2 and β_2 domains, the ones nearest to the cell membrane, assume the characteristic Ig fold, while the α_1 and β_1 domains mimic the class I α_1 and α_2 in forming a groove bounded by two α -helices and a β -pleated sheet floor (figure 4.11a).

The organization of the genes encoding the α chain of the human class II molecule HLA-DR and the main regulatory sequences which control their transcription are shown in figure 4.12.

Several immune response-related genes contribute to the remaining class III region of the MHC

A variety of other genes which congregate within the MHC chromosome region are grouped under the heading of class III. Broadly, one could say that many are directly or indirectly related to

immune defense functions. A notable cluster involves four genes coding for complement components, two of which are for the C4 isotypes C4A and C4B and the other two for C2 and factor B. The cytokines, tumor necrosis factor (TNF, sometimes referred to as TNF α) and lymphotoxin (LT α and LT β), are encoded under the class III umbrella as are three members of the human 70 kDa heat-shock proteins. As ever, things don't quite fit into the nice little boxes we would like to put them in. Even if it were crystal clear where one region of the MHC ends and another begins (and it isn't), some genes located in the middle of the 'classical' (cf. figure 4.13) class I or II regions should more correctly be classified as part of the class III cohort. For example, the *LMP* and *TAP* genes concerned with the intracellular processing and transport of T-cell epitope peptides are found in the class II region (see below), but do not have the classical class II structure nor are they expressed on the cell surface.

Gene map of the MHC

The complete sequence of a human MHC was published at the very end of the last millennium after a gargantuan collaborative effort involving groups in England, France, Japan and the USA. The entire sequence, which represents a composite of different MHC haplotypes, comprises 224 gene loci. Of the 128 of these genes which are predicted to be expressed, it is estimated that about 40% of them have functions related to the immune system. The region between class II and class I in the human contains 60 or so class III genes. An overall view of the main clusters of class I, II

HUMAN	HLA GENE	MICB	MICA	B	C	E	A	G	F
	GENE PRODUCT	MICB	MICA	HLA-B	HLA-C	HLA-E	HLA-A	HLA-G	HLA-F

MOUSE	H-2 GENE	TAPASIN	K	D	L	Q	T	M
	GENE PRODUCT	TAPASIN	H-2K	H-2D	H-2L	Q	T	H-2M

Figure 4.13. MHC class I gene map. The ‘classical’ polymorphic class I genes, *HLA-A*, *-B*, *-C* in humans and *H-2K*, *-D*, *-L* in mice, are highlighted with orange shading and encode peptide chains which, together with β_2 -microglobulin, form the complete class I molecules originally identified in earlier studies as antigens by the antibodies they evoked on grafting into another member of the same species. Note that only some strains of mice possess an *H-2L* gene. The genes expressed most abundantly are *HLA-A* and *-B* in the human and *H-2K* and *-D* in the mouse. The other class I genes (‘class Ib’) are termed ‘nonclassical’ or ‘class I chain-related’. They are oligo- rather than

polymorphic or sometimes invariant, and many are silent or pseudogenes. In the mouse there are approximately 15 *Q* (also referred to as *Qa*) genes, 25 *T* (also referred to as *TL* or *Tla*) genes and 10 *M* genes. *MICA* and *MICB* are ligands for NK cell receptors. Tapasin is involved in peptide transport (cf. p. 94). The gene encoding this molecule is at the centromeric end of the MHC region and therefore is shown in this gene map with respect to the mouse, but in figure 4.14, the class II gene map with respect to the human — look at figure M4.2.1 to see why.

HUMAN	HLA GENE	TAPASIN	DPB	DPA	DOA	DMA	DMB	LMP2	TAP1	LMP7	TAP2	DOB	DQB	DQA	DRB	DRA
	GENE PRODUCT	TAPASIN	DP β	DP α	DO α	DM α	DM β	↓ PROTEASOME COMPLEX		↓ PEPTIDE TRANSPORTER		DO β	DQ β	DQ α	DR β	DR α
			HLA-DP		HLA-DO	HLA-DM						HLA-DO	HLA-DQ		HLA-DR	

MOUSE	H-2 GENE	O α	Ma	Mb2	Mb1	LMP2	TAP2	LMP7	TAP1	Ob	Ab	A α	E β	E α
	GENE PRODUCT	O α	DM α	DM β 2	DM β 1	↓ PROTEASOME COMPLEX		↓ PEPTIDE TRANSPORTER		β	A β	A α	E β	E α
		H-2O	H-2DM							H-2O	H-2A		H-2E	

Figure 4.14. MHC class II gene map with ‘classical’ *HLA-DP*, *-DQ*, *-DR* in the human and *H-2A* (*I-A*) and *H-2E* (*I-E*) in mice more heavily shaded. Both α and β chains of the class II heterodimer are transcribed from closely located genes. There are usually two expressed *DRB* genes, *DRB1* and one of either *DRB3*, *DRB4* or *DRB5*. A similar situation of a single α chain pairing with different β chains is found in the mouse *I-E* molecule. The *LMP2* and *LMP7* genes encode part of the proteasome complex which cleaves cytosolic proteins into small peptides which are transported by the *TAP* gene products into the endoplasmic reticulum. *HLA-DMA* and *-DMB* (mouse *H-2DMA*, *-DMb1* and *-DMb2*) encode the DM $\alpha\beta$ heterodimer which removes

class II-associated invariant chain peptide (CLIP) from classical class II molecules to permit the binding of high affinity peptides. The mouse *H-2DM* molecules are often referred to as *H-2M1* and *H-2M2*, although this is a horribly confusing designation because the term *H-2M* is also used for a completely different set of genes which lie distal to the *H-2T* region and encode members of the class Ib family (cf. figure 4.13). The *HLA-DOA* (alternatively called *HLA-DNA*) and *-DOB* genes (*H-2Oa* and *-Ob* in the mouse) also encode an $\alpha\beta$ heterodimer which may play a role in peptide selection or exchange with classical class II molecules.

and III genes in the MHC of the mouse and human may be gained from figure M4.2.1 in Milestone 4.2. More detailed maps of each region are provided in figures 4.13–4.15. A number of pseudogenes have been omitted from these gene maps in the interest of simplicity.

The cell surface class I molecule based on a trans-membrane chain with three extracellular domains associated with β_2 -microglobulin has clearly proved to be an advantageous model for evolution to mould as evidenced by the variety of molecular species which utilize this structure. It is helpful to subdivide them, first into the **classical class I molecules** (sometimes referred to as class Ia), *HLA-A*, *-B* and *-C* in the human

and *H-2K*, *-D* and *-L* in the mouse. These were defined serologically by the antibodies arising in grafted individuals using methods developed from Gorer’s pioneering studies (Milestone 4.2). Other molecules, sometimes referred to as class Ib, have related structures and are either encoded within the MHC locus itself (‘**nonclassical**’ MHC molecules, for example the human *HLA-E*, *-F* and *-G*, *HFE*, *MICA* and *MICB*, the murine *H-2T*, *-Q* and *-M*), or elsewhere in the genome (‘**class I chain-related**’, including the *CD1* family and *FcRn*). Nonclassical MHC genes are far less polymorphic than the classical MHC, are often invariant, and many are pseudogenes.

HUMAN	CYP21B	C4B	CYP21A	C4A	BF	C2	HSPA1B	HSPA1A	HSPA1L	LTB	TNF	LTA
MOUSE	CYP21A1	C4	CYP21A2	Slp	BF	C2	HSP70-1	HSP70-3	Hsc70t	LTB	TNF	LTA

Figure 4.15. MHC class III gene map. This region is something of a 'rag bag'. Aside from immunologically 'respectable' products like C2, C4, factor B (encoded by the *BF* gene), tumor necrosis factor (*TNF*), lymphotoxin- α and lymphotoxin- β (encoded by *LTA* and *LTB*, respectively) and three 70kDa heat-shock proteins (the *HSPA1A*, *HSPA1B* and *HSPA1L* genes in humans, *HSP70-1*, *HSP70-3* and *Hsc70t* genes in mice), genes not shown in this figure but nonetheless present in this locus include those encoding valyl tRNA synthetase (*G7a*), NOTCH4, which has a number of regulatory activ-

ities, and tenascin, an extracellular matrix protein. Of course many genes may have drifted to this location during the long passage of evolutionary time without necessarily having to act in concert with their neighbors to subserve some integrated defensive function. The 21-hydroxylases (21OHA and B, encoded by *CYP21A* and *CYP21B*, respectively) are concerned with the hydroxylation of steroids such as cortisone. *Slp* (sex-limited protein) encodes a murine allele of C4, expressed under the influence of testosterone.

The genes of the MHC display remarkable polymorphism

Unlike the immunoglobulin system where, as we have seen, variability is achieved in each individual by a **multigenic** system, the MHC has evolved in terms of variability between individuals with a highly **polymorphic** (literally 'many shaped') system based on **multiple alleles** (i.e. alternative genes at each locus). The class I and class II genes are the most polymorphic genes in the human genome; for some of these genes over 200 allelic variants have been identified. Class I HLA-A, -B and -C molecules are highly polymorphic and so are the class II β chains (HLA-DR β most, -DP β next and -DQ β third) and, albeit to a lesser extent than the β chains, the α chains of -DP and -DQ. HLA-DR α and β_2 -microglobulin are invariant in structure. The amino acid changes responsible for this polymorphism are restricted to the α_1 and α_2 domains of class I and to the α_1 and β_1 domains of class II. It is of enormous significance that they occur essentially in the β -sheet floor and on the inner surfaces of the α -helices which line the central cavity (figure 4.11a) and also on the upper surfaces of the helices.

The MHC region represents an outstanding hotspot with mutation rates two orders of magnitude higher than non-MHC loci. These multiple allelic forms can be generated by a variety of mechanisms: point mutations, recombination, homologous but unequal crossing over and **gene conversion**.

The degree of sequence homology and an increased occurrence of the dinucleotide motif 5'-cytosine-guanine-3' (to produce what are referred to as CpG islands) seem to be important for gene conversion, and it has been suggested that this might involve a DNA-nicking activity which targets CpG-rich DNA sequences. MHC genes that lack these sequences, for example H-2Ea^d and HLA-DRA, do not appear

to undergo gene conversion, whereas those that possess CpG islands act as either donors (e.g. H-2Eb^b, H-2Q2^k, H-2Q10^b), acceptors (e.g. H-2A^b) or both (e.g. H-2K^k, HLA-DQB1). The large number of pseudogenes within the MHC may represent a stockpile of genetic information for the generation of polymorphic diversity in the 'working' class I and class II molecules.

Nomenclature

Since much of the experimental work relating to the MHC is based on experiments in our little laboratory friend, the mouse, it may be helpful to explain the nomenclature used to describe the allelic genes and their products. If someone says to you in an obscure language 'we are having free elections', you fail to understand, not because the idea is complicated but because you do not comprehend the language. It is much the same with the shorthand used to describe the H-2 system which looks unnecessarily frightening to the uninitiated. In order to identify and compare allelic genes within the H-2 complex in different strains, it is usual to start with certain pure homozygous inbred strains, obtained by successive brother-sister matings, to provide the prototypes. The collection of genes in the H-2 complex is called the **haplotype** and the haplotype of each prototypic inbred strain will be allotted a given superscript. For example, the DBA strain haplotype is designated H-2^d and the genes constituting the complex are therefore H-2K^d, H-2Aa^d, H-2Ab^d, H-2D^d and so on; their products will be H-2K^d, H-2A^d and H-2D^d and so forth (figure 4.16). When new strains are derived from these by genetic recombination during breeding, they are assigned new haplotypes, but the individual genes are designated by the haplotype of the prototype strain from which they were derived. Thus the A/J strain produced by genetic cross-over during interbreeding between (H-2^k × H-2^d) F1 mice

Strain	Haplotype	MHC Designation	I	II				III	I	
C57BL	<i>b</i>	<i>H-2^b</i>	<i>K^b</i>	<i>Ab^b</i>	<i>Aa^b</i>	<i>Eb^b</i>	<i>Ea^b</i>	<i>C4^b</i>	<i>etc</i>	<i>D^b</i> <i>etc</i>
CBA	<i>k</i>	<i>H-2^k</i>	<i>K^k</i>	<i>Ab^k</i>	<i>Aa^k</i>	<i>Eb^k</i>	<i>Ea^k</i>	<i>C4^k</i>	<i>etc</i>	<i>D^k</i> <i>etc</i>

Figure 4.16. How the definition of *H-2* haplotype works. Pure strain mice homozygous for the whole *H-2* region through prolonged brother–sister mating for at least 20 generations are each arbitrarily assigned a **haplotype** designated by a superscript. Thus the particular set of alleles which happens to occur in the strain named C57BL is assigned the haplotype *H-2^b* and the particular nucleotide

sequence of each allele in its MHC is labeled as **gene^b**, e.g. *H-2K^b*, etc. It is obviously more convenient to describe a given allele by the haplotype than to trot out its whole nucleotide sequence, and it is easier to follow the reactions of cells of known *H-2* make-up by using the haplotype terminology—see, for example, the interpretation of the experiment in figure 4.17.

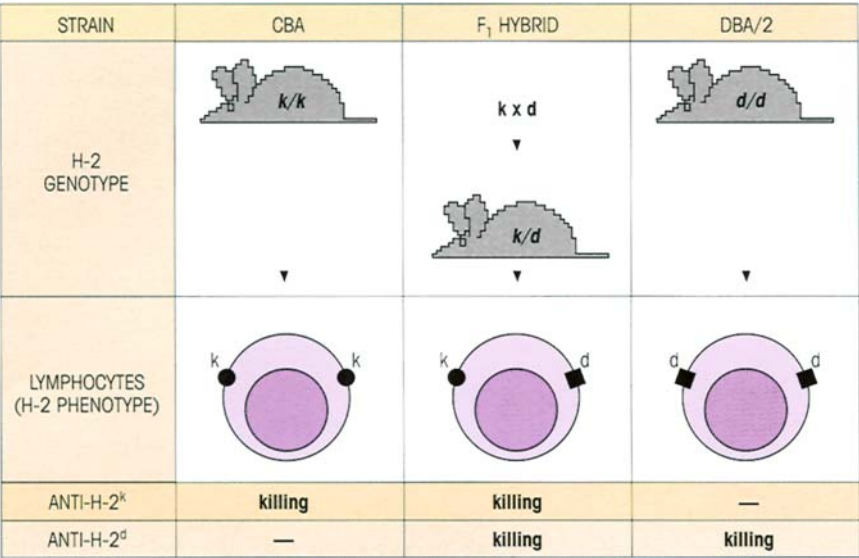


Figure 4.17. Inheritance and codominant expression of MHC genes. Each homozygous (pure) parental strain animal has two identical chromosomes bearing the *H-2* haplotype, one paternal and the other maternal. Thus in the present example we designate a strain which is *H-2^k* as *k/k*. The first familial generation (F1) obtained by crossing the pure parental strains CBA (*H-2^k*) and DBA/2 (*H-2^d*) has the *H-2* genotype *k/d*. Since 100% of F1 lymphocytes are killed in the presence of complement by antibodies to *H-2^k* or to *H-2^d* (raised by injecting *H-2^k* lymphocytes into an *H-2^d* animal and vice versa), the MHC molecules encoded by both parental genes must be expressed on every lymphocyte. The same holds true for other tissues in the body.

Table 4.2. The haplotypes of the *H-2* complex of some commonly used mouse strains and recombinants derived from them. A/J was derived by interbreeding (*k* × *d*) F1 mice, recombination occurring between E (class II) and S (class III) regions*.

STRAIN	HAPLOTYPE	ORIGIN OF INDIVIDUAL REGIONS				
		K	A	E	S	D
C57BL	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>
CBA	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>
DBA/2	<i>d</i>	<i>d</i>	<i>d</i>	<i>d</i>	<i>d</i>	<i>d</i>
A/J	<i>a</i>	<i>k</i>	<i>k</i>	<i>k*</i>	<i>d</i>	<i>d</i>
B.10A(4R)	<i>h4</i>	<i>k</i>	<i>k</i>	<i>b</i>	<i>b</i>	<i>b</i>

(figure 4.17) is arbitrarily assigned the haplotype *H-2^a*, but table 4.2 shows that individual genes in the complex are identified by the haplotype symbol of the original parents.

Inheritance of the MHC

Pure strain mice derived by prolonged brother–sister mating are homozygous for each pair of homologous

chromosomes. Thus, in the present context, the haplotype of the MHC derived from the mother will be identical to that from the father; animals of the C57BL strain, for example, will each bear two chromosomes with the *H-2^b* haplotype (cf. table 4.2).

Let us see how the MHC behaves when we cross two pure strains of haplotypes *H-2^k* and *H-2^d*, respectively. We find that the lymphocytes of the offspring (the F1 generation) all display *both* *H-2^k* and *H-2^d* molecules on their surface, i.e. there is **codominant expression** (figure 4.17). If we go further and breed F1s together, the progeny have the genotypes *k, k/d* and *d* in the proportions to be expected if the **haplotype segregates as a single Mendelian trait**. This happens because the *H-2* complex spans 0.5 centimorgans, equivalent to a recombination frequency between the *K* and *D* ends of 0.5%, and the haplotype tends to be inherited *en bloc*. Only the relatively infrequent recombinations caused by meiotic cross-over events, as described for the A/J strain above, reveal the complexity of the system.

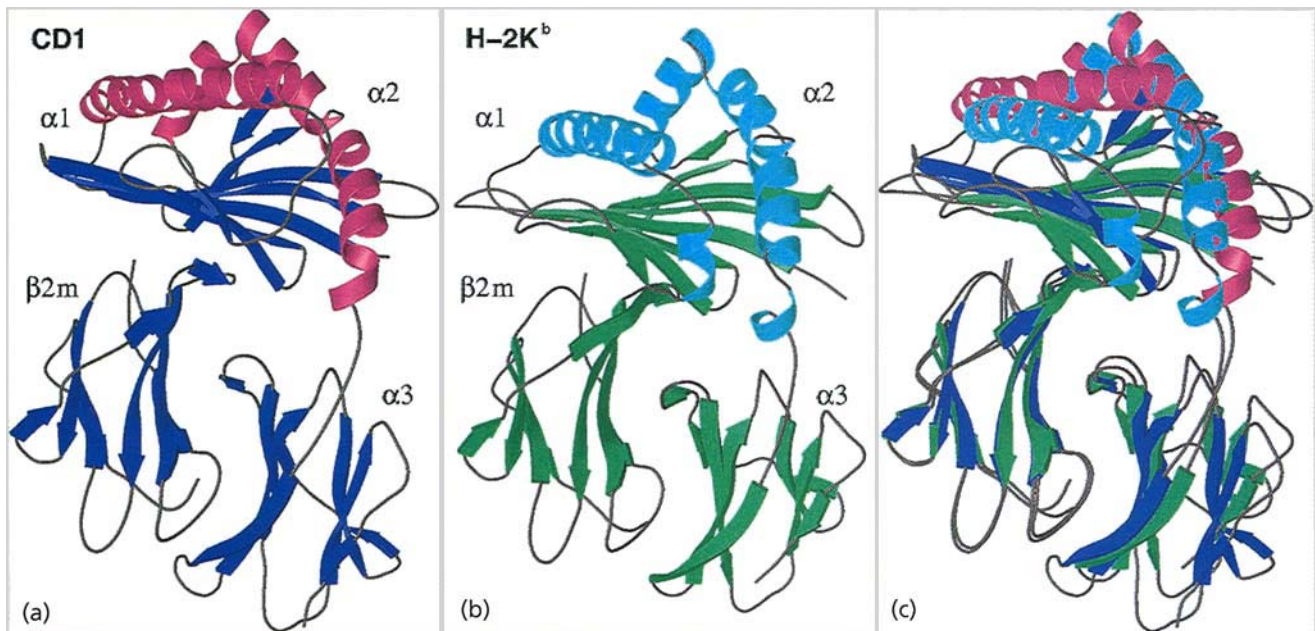


Figure 4.18. Comparison of the crystal structures of CD1 and MHC class I. (a) Backbone ribbon diagram of mouse CD1d1 (red, α -helices; blue, β -strands). (b) Ribbon diagram of the mouse MHC class I molecule H-2K^b (cyan, α -helices; green, β -strands). (c) Superposition using alignment of β_2 -microglobulin highlights some of the differences between CD1d1 and H-2K^b. Note in particular the shifting of the α -helices. This produces a deeper and more voluminous groove in CD1d1, which is narrower at its entrance compared with H-2K^b. (Reprinted with permission from Porcelli S.A. *et al.* (1998) *Immunology Today* 19, 362.)

The tissue distribution of MHC molecules

Essentially, all nucleated cells carry classical class I molecules. These are abundantly expressed on both lymphoid and myeloid cells, less so on liver, lung and kidney and only sparsely on brain and skeletal muscle. In the human, the surface of the placental extravillous cytotrophoblast lacks HLA-A and -B, although there is now some evidence that it may express HLA-C. What is well established is that the extravillous cytotrophoblast and other placental tissues bear HLA-G, a molecule which generally lacks allodeterminants and which does not appear on most other body cells, except for medullary and subcapsular epithelium in the thymus, and on blood monocytes following activation with γ -interferon. The role of HLA-G in the placenta is unclear, but it may function as a replacement for allodeterminant-bearing classical class I molecules and/or may play a role in shifting potentially harmful Th1 responses towards a Th2-type response. Class II molecules, on the other hand, are highly restricted in their expression, being present only on B-cells, dendritic cells, macrophages and thymic epithelium.

However, when activated by agents such as γ -interferon, capillary endothelia and many epithelial cells in tissues other than the thymus express surface class II and increased levels of class I.

MHC function

Although originally discovered through transplantation reactions, as we will see in subsequent chapters, the MHC molecules act as cell surface markers which enable infected cells to signal cytotoxic and helper T-cells. There is no doubt that this role in immune responsiveness is immensely important, and in this respect the rich **polymorphism of the MHC** region represents a species response to **maximize protection against diverse microorganisms**. An apparent example is the malaria-driven selection at the HLA-B locus whereby resistance to severe malaria resulting from strains of *Plasmodium falciparum* in East Africa is associated with HLA-DRB1*0101, whereas HLA-DRB1*1302 confers resistance to West African strains of the parasite.

The nonclassical MHC and class I chain-related molecules

These molecules include the CD1 family which utilize β_2 -microglobulin and have a similar overall structure to the classical class I molecules (figure 4.18). They are, however, encoded by a set of genes on a different chromosome to the MHC, namely on chromosome 1 in humans and chromosome 3 in the mouse. Like its true

MHC counterparts, CD1 is involved in the presentation of antigens to T-cells, but the antigen-binding groove is to some extent covered over, contains mainly hydrophobic amino acids, and is accessible only through a narrow entrance. Instead of binding peptide antigens, the CD1 molecules generally present lipids or glycolipids. At least four different CD1 molecules are found expressed on human cells; CD1a, b and c are present on cortical thymocytes, dendritic cells and a subset of B-cells, whilst CD1d is expressed on intestinal epithelium, hepatocytes and all lymphoid and myeloid cells. Mice appear to only express two different CD1 molecules which are both similar to the human CD1d in structure and tissue distribution and are referred to as CD1d1 and CD1d2 (or CD1.1 and CD1.2).

Genes in the MHC itself which encode nonclassical MHC molecules include the H-2T, -Q and -M loci in mice, each of which encodes a number of different molecules. The T22 and T10 molecules, for example, are induced by cellular activation and are recognized directly by $\gamma\delta$ TCR without a requirement for antigen, possibly suggesting that they are involved in triggering immunoregulatory $\gamma\delta$ T-cells. Other nonclassical class I molecules do bind peptides, such as H-2M3 which presents *N*-formylated peptides produced either in mitochondria or by bacteria.

In the human, HLA-E binds a nine-amino acid peptide derived from the signal sequence of HLA-A,

-B, -C and -G molecules, and is recognized by the CD94/NKG2 receptors on NK cells and cytotoxic T-cells, as well as by the $\alpha\beta$ TCR on some cytotoxic T-cells. HLA-E is upregulated when other HLA alleles provide the appropriate leader peptides, thereby perhaps allowing NK cells to monitor the expression of polymorphic class I molecules using a single receptor. The murine homolog, Qa-1, has a similar function.

The stress-inducible MICA and MICB (MHC class I chain-related molecules) have the same domain structure as classical class I and display a relatively high level of polymorphism. They are present on epithelial cells, mainly in the gastrointestinal tract and in the thymic cortex, and are recognized by the NKG2D activating molecule. One possible role for this interaction is in the promotion of NK and T-cell antitumor responses.

The function of HLA-F is unclear. In contrast, although HLA-G shows extremely limited polymorphism, it is known to bind a range of self peptides with a defined binding motif and there is evidence for HLA-G restricted T-cells.

HFE, previously referred to as HLA-H, possesses an extremely narrow groove which is unable to bind peptides, and it may serve no role in immune defense. However, it binds to the transferrin receptor and appears to be involved in iron uptake. A point mutation (C282Y) in HFE is found in 70–90% of patients with hereditary hemochromatosis.

SUMMARY

The B-cell surface receptor for antigen

- The B-cell inserts its Ig gene product containing a transmembrane segment into its surface where it acts as a specific receptor for antigen.
- The surface Ig is complexed with the membrane proteins Ig- α and Ig- β which become phosphorylated on cell activation and transduce signals received through the Ig antigen receptor.

The T-cell surface receptor for antigen

- The receptor for antigen is a transmembrane dimer, each chain consisting of two Ig-like domains.
- The outer domains are variable in structure, the inner ones constant, rather like a membrane-bound Fab.
- Both chains are required for antigen recognition.
- Most T-cells express a receptor (TCR) with α and β chains (TCR2). A separate lineage (TCR1) bearing $\gamma\delta$ receptors is transcribed strongly in early thymic ontogeny but is associated mainly with epithelial tissues in the adult.

- The encoding of the TCR is similar to that of immunoglobulins. The variable region coding sequence in the differentiating T-cell is formed by random translocation from clusters of *V*, *D* (for β and δ chains) and *J* segments to give a single recombinant *V(D)J* sequence for each chain.
- Like the Ig chains, each variable region has three hypervariable sequences which function in antigen recognition.
- The CD3 complex, composed of γ , δ , ϵ and either ζ , η or η_2 covalently linked dimers, forms an intimate part of the receptor and has a signal transducing role following ligand binding by the TCR.

The generation of antibody diversity for antigen recognition

- Ig heavy and light chains and TCR α and β chains generally are represented in the germ-line by between 30 and 75 variable region genes, between 2 and 25 *D* segment minigenes (Ig heavy and TCR β and δ only) and 4–60 short *J* segments.

(continued)

- TCR γ and δ chains are encoded by far fewer genes.
- Random recombination of any single V , D and J from each gene cluster generates approximately 7.5×10^3 Ig heavy chain VDJ sequences, 200 light chains, 4.5×10^3 TCR α , 1×10^3 TCR β , but only 60 TCR γ and 72 TCR δ .
- Random interchain combination produces roughly 1.5×10^6 Ig, 4.5×10^6 TCR $\alpha\beta$ and 4.3×10^3 TCR $\gamma\delta$ receptors.
- Further diversity is introduced at the junctions between V , D and J segments by variable combination as they are spliced together by recombinase enzymes and by the N-region insertion of random nontemplated nucleotide sequences. These mechanisms may be particularly important in augmenting the number of specificities which can be squeezed out of the relatively small $\gamma\delta$ pool.
- Useless or self-reactive receptors can be replaced by receptor editing.
- In addition, after a primary response, B-cells but not T-cells undergo high rate somatic mutation affecting the V regions.

NK receptors

- NK cells bear a number of receptors with Ig-type domains and other receptors with C-type lectin domains. Members of both types of receptor family can function as inhibitory or activatory receptors to determine if the target cell is killed.

MHC

- Each vertebrate species has an MHC identified originally through its ability to evoke very powerful transplantation rejection.
- Each contains three classes of genes. Class I encodes 44 kDa transmembrane polypeptides associated at the cell

surface with β_2 -microglobulin. Class II molecules are transmembrane heterodimers. Class III products are heterogeneous but include complement components linked to the formation of C3 convertases, heat-shock proteins and tumor necrosis factors.

- The genes display remarkable polymorphism. A given MHC gene cluster is referred to as a 'haplotype' and is usually inherited *en bloc* as a single Mendelian trait, although its constituent genes have been revealed by cross-over recombination events.
- Classical class I molecules are present on virtually all cells in the body and signal cytotoxic T-cells.
- Class II molecules are particularly associated with B-cells, dendritic cells and macrophages but can be induced on capillary endothelial cells and epithelial cells by γ -interferon. They signal T-helpers for B-cells and macrophages.
- The two domains distal to the cell membrane form a peptide binding cavity bounded by two parallel α -helices sitting on a floor of β -sheet strands; the walls and floor of the cavity and the upper surface of the helices are the sites of maximum polymorphic amino acid substitutions.
- Silent class I genes may increase polymorphism by gene conversion mechanisms.
- Nonclassical MHC molecules and MHC-like molecules have a number of functions, and include CD1 which presents lipid and glycolipid antigens to T-cells, and HLA-E which presents signal sequence peptides from classical class I molecules to the CD94/NKG2 receptor of NK cells.

See the accompanying website (www.roitt.com) for multiple choice questions.

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The primary interaction with antigen

Introduction, 80

The nature of B-cell epitopes, 81

Of epitopes and antigen determinants, 81

Identification of B-cell epitopes, 82

Antigens and antibodies interact by spatial complementarity not by covalent bonding, 83

Variation in hapten structure shows importance of shape, 83

Spatial complementarity of epitope and paratope can be demonstrated, 83

Antigen–antibody bonds are readily reversible, 84

The forces binding antigen to antibody become large as intermolecular distances become small, 85

Affinity measures strength of binding of antigen and antibody, 85

The avidity of antiserum for antigen — the bonus effect of multivalency, 87

The specificity of antigen recognition by antibody is not absolute, 89

What the T-cell sees, 90

Haplotype restriction reveals the need for MHC participation, 91

T-cells recognize a linear peptide sequence from the antigen, 91

Processing of intracellular antigen for presentation by class I MHC, 93

Processing of antigen for class II MHC presentation follows a different pathway, 95

The nature of the 'groovy' peptide, 96

Binding to MHC class I, 97

Binding to MHC class II, 97

The $\alpha\beta$ T-cell receptor forms a ternary complex with MHC and antigenic peptide, 99

Topology of the ternary complex, 99

T-cells with a different outlook, 100

Nonclassical class I molecules can also present antigen, 100

Some T-cells have NK markers, 102

$\gamma\delta$ TCRs have some features of antibody, 102

Superantigens stimulate whole families of lymphocyte receptors, 103

Bacterial toxins represent one major group of T-cell superantigens, 103

Endogenous mouse mammary tumor viruses (MMTV) act as superantigens, 103

Microbes can also provide B-cell superantigens, 104

The recognition of different forms of antigen by B- and T-cells is advantageous to the host, 104

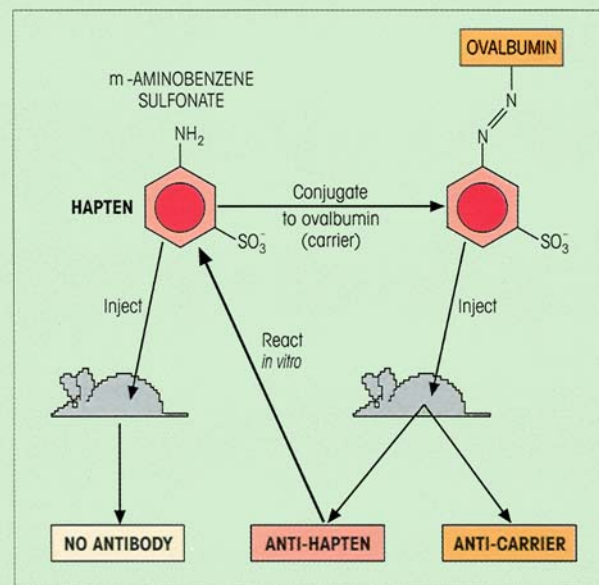
Summary, 105

Further reading, 106

INTRODUCTION

A man cannot be a husband without a wife and a molecule cannot be an antigen without a corresponding antiserum or antibody or T-cell receptor. The term **antigen** is used in two senses, the first to describe a molecule which generates an immune response (also called an **immunogen**) and the second, a molecule which reacts with antibodies or primed T-cells irrespective of its ability to generate them. If this last situation sounds a trifle confusing, an example may help. A mouse injected with its own red cells, not too surprisingly, will not make any antibodies; if it is now given rat erythrocytes,

Figure 5.1. A hapten on its own will not induce antibodies. However, it will react *in vitro* with antibodies formed to a conjugate with an immunogenic carrier.



(continued)

antibodies are formed to both rat and *mouse* red cells and the latter bind to the animal's own cells *in vivo*, i.e. the mouse red cell acts as antigen in binding antibodies even though unable to evoke their formation. Similarly, **haptens**, which are small well-defined chemical groupings such as dinitrophenyl (DNP; cf. p. 37) or *m*-aminobenzene sulfonate, are not immunogenic on their own but will react with preformed antibodies induced by injection of the hapten linked to a 'carrier' molecule which is itself an immunogen (figure 5.1).

The parts of the hypervariable regions on the antibody which contact the antigen are termed the **paratope** and the part of the antigen which is in contact with the paratope is designated the **epitope**. To get some idea of size, if the antigen is a linear peptide or carbohydrate, the combining site can usually accommodate up to five or six amino acid residues or hexose units. With a globular protein, typically 14–21 amino acid residues are in contact on both the antibody and the antigen (cf. figure 5.5).

THE NATURE OF B-CELL EPITOPES

Of epitopes and antigen determinants

Antibodies formed in response to immunization with a native globular protein (as distinct from a fibrillar protein) do not tend to react well with denatured preparations, and this is consistent with the view that the majority recognize topographic (surface) structures (i.e. epitopes) which depend upon the conformation of the native molecule. For this reason, antibodies to native proteins do not usually react as strongly with peptides having the same primary sequence (figure 5.2). When individual epitopes are mapped using homogeneous monoclonal antibodies (cf. p. 120), they are frequently seen to involve amino acid residues far apart in the primary sequence, but brought together by the folding of the peptide chains in the native protein (figures 5.3 and 5.5). It seems reasonable to talk of **discontinuous** or assembled rather than **continuous** or sequential epitopes in these cases.

If one were to take each individual antibody within an antiserum raised to a protein antigen and plot the approximate center of the corresponding epitope on

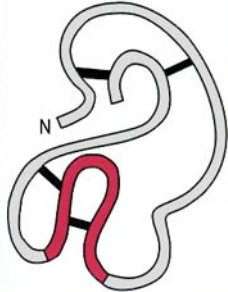
	LYSOZYME	ISOLATED LOOP PEPTIDE	REDUCED LOOP PEPTIDE
			
Anti-lysozyme	++	+	–
Anti-loop peptide	+	++	–

Figure 5.2. Specificity and three-dimensional configuration in a globular protein, lysozyme. Antibodies to the whole molecule and to the isolated loop peptide do not react with the peptide after reduction of its disulfide bond, showing that the linear reduced peptide has lost the antigenic configuration it had when held as a loop even though the amino acid sequence is unchanged. (From Maron E., Shiowa C., Arnon R. & Sela M. (1971) *Biochemistry* 10, 763.)

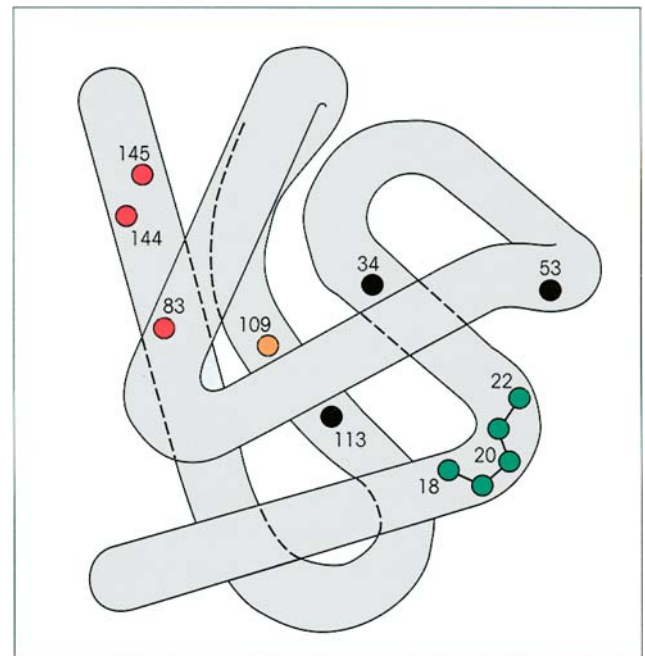


Figure 5.3. Epitope residues on the folded peptide chain of sperm whale myoglobin. Amino acid residues 34, 53 and 113 (●) contribute to the epitope recognized by one homogeneous monoclonal antibody, 83, 144 and 145 (●), to another. These are clearly discontinuous epitopes. Amino acids 18–22 (●●●) are postulated to form part of a continuous epitope based on reactions with the isolated peptides. Much of the myoglobin chain is in the α -helical form. Residue 109 (●) is critical for T-cell recognition and so far no antibodies reacting with this site have been demonstrated. (Based on Benjamin D.C. *et al.* (1986) *Annual Review of Immunology* 2, 67.)

the antigen surface, one would almost certainly finish up with a 'contour map' of epitope density indicating regions on the antigen surface of **dominant epitope clusters** (figure 5.4a and b). Each of these clusters is as near as we can get to defining an antigen **determinant**. It is important to be aware that each antigen usually bears several determinants on its surface, which may well be structurally distinct from each other; thus a monoclonal antibody reacting with one determinant will usually not react with any other determinants on the same antigen unless the molecule has axes of symmetry (figure 5.4c).

Identification of B-cell epitopes

This is a subject of particular interest to those wishing to make simple peptide substitutes for complex protein antigens. In general, large proteins, because they have more potential determinants, are better antigens than small ones. The more foreign an antigen, that is the less similar to self configurations which induce tolerance, the more effective it is in provoking an immune response.

Parts of the peptide chains which protrude significantly from the globular surface tend to be sites of high epitope density. The least antigenic segments of the surface are associated with neighboring concave regions containing water molecules which may be more difficult to displace. However, prediction of B-cell epitopes, even when one knows the three-dimensional structure of an antigen, is still a pretty hopeless task, not least because each immunized host has its own way of recognizing the different regions of a given antigen. In fact, the only recognition unit that can be deployed to identify an epitope is that which defines it, namely the antibody bearing the complementary paratope.

The greatest precision is undoubtedly provided by X-ray crystallographic analysis of a complex of a monoclonal antibody (or fragment thereof) with the antigen (cf. figure 5.5). Whilst over 100 such structures have been solved, this approach is not for the run of the mill scientist, being difficult, time-consuming and very 'high tech'. An alternative strategy, applicable to protein antigens, is to carry out a series of mutations to locate the residues which provide the dominant binding to antibody — useful but not without its limitations. Attempts to mimic the epitope by building synthetic peptides will be described later (p. 292). To anticipate, this is good for linear epitopes and frustrating for those which are discontinuous.

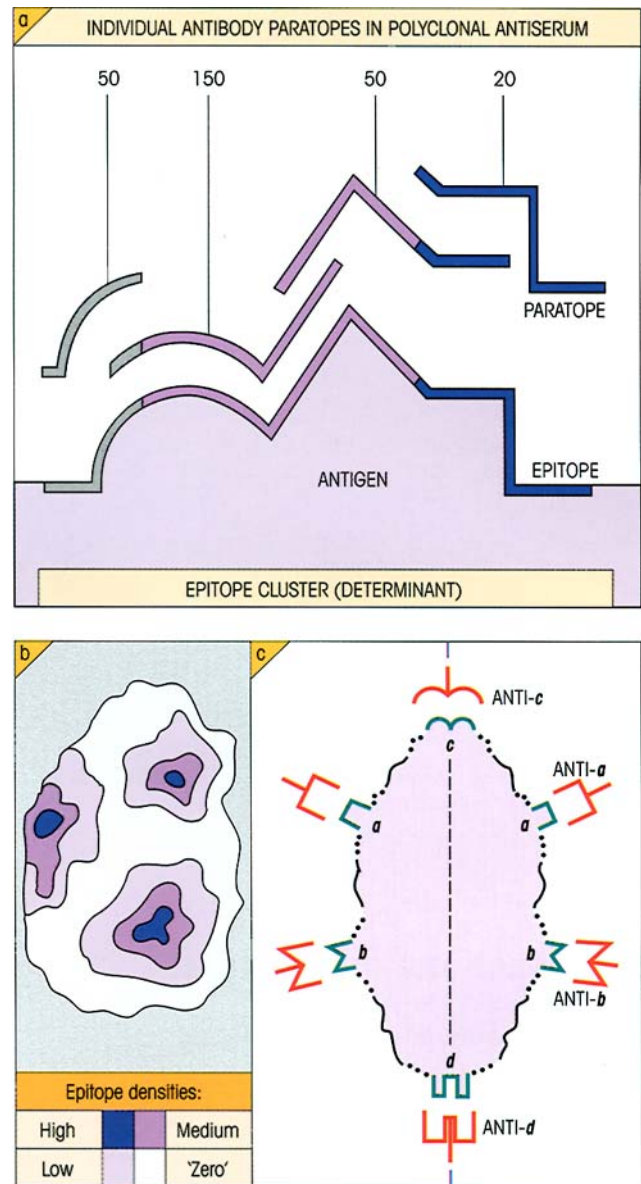


Figure 5.4. A globular protein antigen usually bears a mosaic of determinants (dominant epitope clusters) on its surface, defined by the heterogeneous population of antibody molecules in a given antiserum. (a) Highly idealized diagram illustrating the idea that individual antibodies in a polyclonal antiserum with different combining sites (paratopes) can react with overlapping epitopes forming a determinant on the surface of the antigen. The numbers refer to the imagined relative frequency of each antibody specificity. (b) Hypothetical 'contour' map of surface showing how determinants represent regions of clustering but overlapping epitopes, whose positions are plotted as the center of the area making contact with antibody. The actual size of a single epitope may be gauged by looking at figure 5.5. (c) Cross-section of a theoretical antigen with an axis of symmetry displaying six determinants including two pairs which are identical. The clusters of overlapping antibodies to each determinant (one representative of each antibody cluster is shown) do not react with the other structurally unrelated determinants.

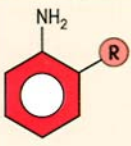
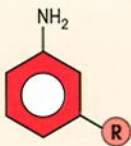
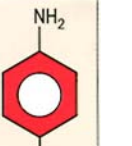
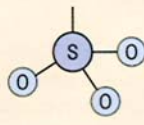
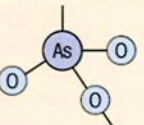
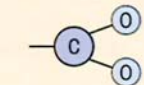
ANTIGENS AND ANTIBODIES INTERACT BY SPATIAL COMPLEMENTARITY NOT BY COVALENT BONDING

Variation in hapten structure shows importance of shape

Once a method had been found for raising antibodies to small chemically defined haptens (figure 5.1), it then became possible to relate variations in the chemical structure of a hapten to its ability to bind to a given antibody. In one experiment, antibodies raised to *m*-aminobenzene sulfonate were tested for their ability to combine with *ortho*, *meta* and *para* isomers of the hapten and related molecules in which the sulfonate group was substituted by arsonate or carboxylate (table 5.1). The hapten with the sulfonate group in the *ortho* position combines somewhat less well with the antibody

than the original *meta* isomer, but the *para*-substituted compound (chemically similar to the *ortho*) shows very poor reactivity. The substitution of arsonate for sulfonate leads to weaker combination with the antibody; both groups are negatively charged and have a tetrahedral structure but the arsonate group is larger in size and has an extra H atom. The aminobenzoates in which the sulfonate is substituted by the negatively charged but planar carboxylate group show even less affinity for the antibody. It would appear that the **overall** configuration of the hapten is even more important than its **chemical** nature, i.e. the hapten is recognized by the overall three-dimensional shape of its outer electron cloud as distinct from its chemical reactivity. The production of antibodies against such strange moieties as benzene sulfonate and arsonate becomes more comprehensible if they are thought to be directed against a particular electron-cloud shape rather than a specific chemical structure.

Table 5.1. Effect of variations in hapten structure on strength of binding to antibodies raised against *m*-aminobenzene sulfonate. The reaction of the antibody with the original hapten against which it was raised is highlighted by the box. (From Landsteiner K. & van der Scheer J. (1936) *Journal of Experimental Medicine* 63, 325.)

	 ORTHO	 META	 PARA
R = SULFONATE  tetrahedral	++	+++	±
R = ARSONATE  tetrahedral	-	+	-
R = CARBOXYLATE  planar	-	±	-

Spatial complementarity of epitope and paratope can be demonstrated

It eventually proved possible, with not a little difficulty, to crystallize a complex of the Fab fragment of monoclonal antilysozyme with its antigen. X-ray analysis of these crystals was convincing; antigen and antibody fitted strongly together due to complementarity in shape over a wide area of contact (figure 5.5). Similar studies with further monoclonal antibodies reacting with the same antigen confirmed the 'lock and key' type fit between paratope and epitope. It is important not to regard the 'lock and key' as inflexible entities like two pieces of rock, since this might make it very difficult for an animal to produce an antibody with such a unique complementary surface. In fact, most monoclonal antilysozyme antibodies studied in this way revealed significant changes in the polypeptide backbone of up to 1.0 Å as the antibody complexed with its antigen (cf. figure 5.6), and if one adds into the equation the possibility of rotational movement of the amino acid side-chains and alterations in the relative positions of the variable domains of light and heavy chains (V_L/V_H) in the Fab, it would seem more correct to think of antigen and antibody as surfaces which are to some extent mutually deformable—more like clouds than rocks as Lerner has so graphically expressed it. In this context, it is pertinent to draw attention to the association between those parts of a protein antigen which provide the dominant epitopes for antibody binding, and their peptide chain flexibility as measured by the temperature factors derived from X-ray crystallography. This mutual

accommodation of structures on antigen and antibody obviously permits the maximum contact, but there must be a price to pay in that energy has to be expended in inducing these conformational changes. In general, though, this will be more than compensated by the intrinsic free energy change associated with the formation of the new binding sites (figure 5.7a and b). This input of energy to produce conformational change leading to strong binding appears to be the factor behind the frequent occurrence of normally 'buried' hydrophobic side-chains as contact residues for antibody.

With globular protein antigens, the area of contact between epitope and paratope is quite large, of the order of 600–800 Å². Tyrosines and tryptophans tend to occur with greater frequency in the combining site than they do in the remainder of the antibody molecule. Both can form H bonds with solvent or antigen, yet have large hydrophobic surfaces, undeniably useful characteristics for residues exposed to water when the antibody is free but buried (i.e. excluded from solvent contact) when interacting closely with antigen.

Antigen–antibody bonds are readily reversible

If the link between epitope and paratope is entirely dependent on spatial complementarity and does not involve the formation of covalent chemical bonds, it should not be too difficult to pull them apart. This can easily be put to the test. If one puts a mixture of the hapten with antibody inside a dialysis bag, the hapten will be found to diffuse out into the surrounding fluid until an equilibrium is reached in which some hapten is bound to antibody and some is free; if this exterior fluid is continually renewed, all the hapten will be lost from the bag showing that it can be completely dissociated from the antibody (figure 5.8). With larger antigens, the complexes can be dissociated by a change in

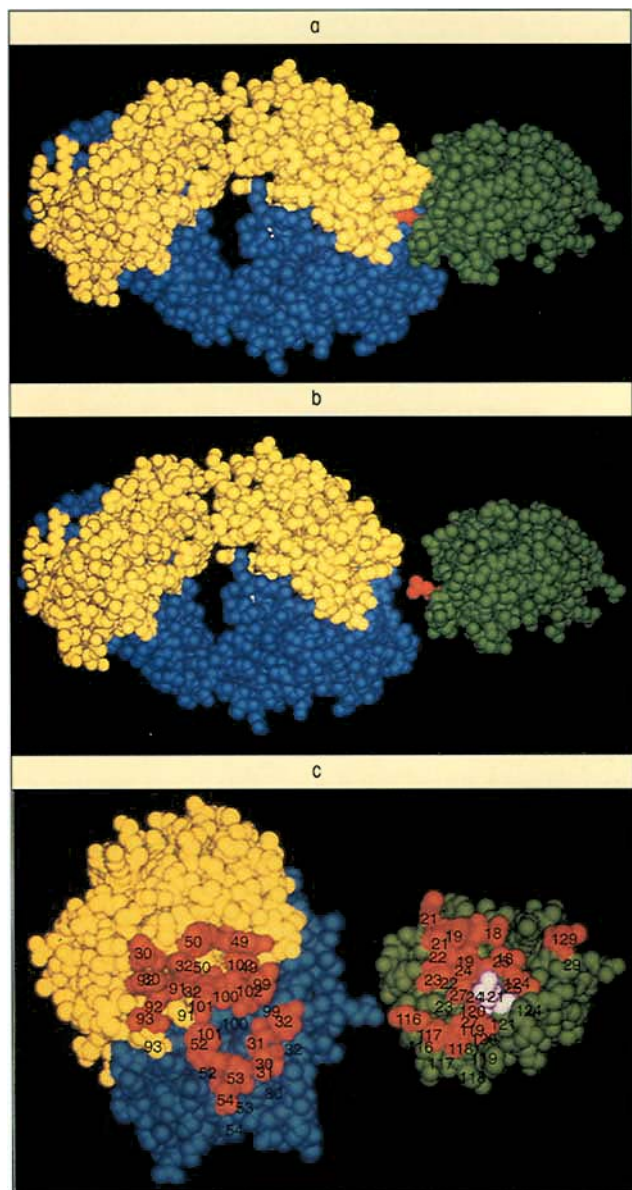


Figure 5.5. Structure of the contact regions between a monoclonal Fab antilysozyme and lysozyme. (a) Space-filling model showing Fab and lysozyme molecules fitting snugly together. Antibody heavy chain, blue; light chain, yellow; lysozyme, green with its glutamine 121 in red. (b) Fab and lysozyme models pulled apart to show how the protuberances and depressions of each are complementary to each other. (c) End-on views of antibody combining site (left) and the lysozyme epitope (right) obtained from (b) by rotating each molecule 90° about a vertical axis. Contact residues are red, except Gln121 in light purple. The Gln121 fits into an antibody surface cavity surrounded by V_L residues 32, 91, 92 and 93 and V_H residue 101. The sixteen contact residues in the lysozyme epitope comprise: eight amino acids from a stretch of 10 amino acids at positions 18–27, seven amino acids from a stretch of nine amino acids at positions 116–124, together with amino acid residue 129, i.e. this is clear evidence for a discontinuous epitope. All the CDRs make contact with the antigen: in the heavy chain residues 30–32 (CDR1), 52–54 (CDR2) and 99–102 (CDR3) and in the light chain residues 30 and 32 (CDR1), 49 and 50 (CDR2) and 91–93 (CDR3). All contacting residues may not contribute positively to the attractive forces between antigen and antibody; the striking influence of the Gln121 is revealed by the poor binding of lysozymes from other species in which the Gln121 is replaced by histidine. (Reproduced with permission from Amit A. *et al.* (1986) *Science* 233, 747. Copyright © 1986 by the AAAS.)

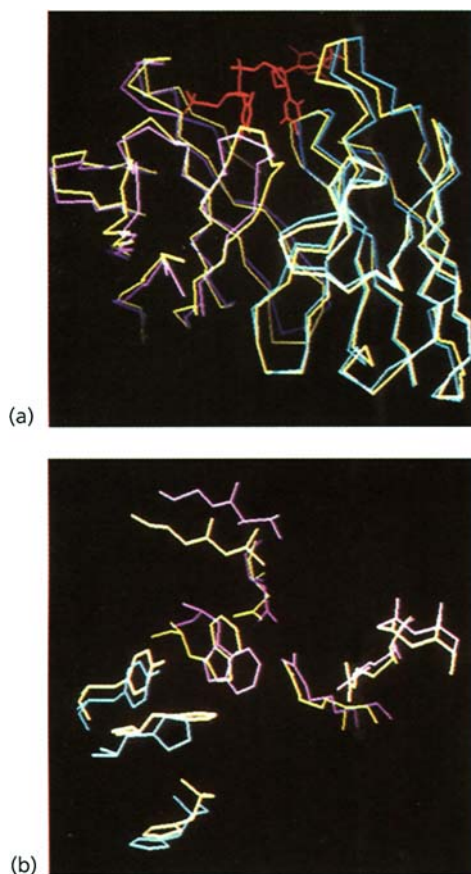


Figure 5.6. Antibody structural flexibility on complexing with the antigen. (a) Significant conformational changes (0.5–1.0 Å) occur in the C α backbone of the complementarity determining regions (CDRs) of an autoantibody Fab specific for single-stranded DNA on complexing with a fragment of antigen (thymidine trimer). Only the Fv (V_H + V_L) domains of the antibody are shown. The unliganded Fv is yellow and the ligand (antigen) red. In the complexed form, the V_L is blue and the V_H purple. (b) Movements in the contacting amino acid side-chains as a result of binding antigen. (Reproduced with permission from Herron J.N., He X.M., Ballard D.W., Blier P.R., Pace P.E., Bothwell A.L.M., Voss E.W. Jr. & Edmundson A.B. (1991) *Proteins* **11**, 150. Reprinted by permission of Wiley-Liss, a division of John Wiley and Sons, Inc.)

pH which brings about alterations in protein conformation and destroys the complementarity of the two reactants. As will be seen subsequently (p. 125), this principle can be used for the purification of either antigens or antibodies by affinity chromatography.

THE FORCES BINDING ANTIGEN TO ANTIBODY BECOME LARGE AS INTERMOLECULAR DISTANCES BECOME SMALL

The forces involved in antibody binding to antigen are noncovalent and therefore reversible, and are gener-

ally only effective over short distances. They may be classified under four headings.

1 Electrostatic. The attraction between oppositely charged ionic groups (figure 5.9a). The opposing electrostatic forces need to approximately match each other if the antibody is to substantially bind the antigen.

2 Hydrogen bonding. The sharing of a hydrogen between two electronegative atoms (figure 5.9b). Hydrogen bonding can occur between atoms such as oxygen, nitrogen or sulfur. In addition, water molecules can fill cavities in the interface which would otherwise have a destabilizing effect on the complex and these water molecules can form hydrogen bonds to bridge the antibody and antigen.

3 Hydrophobic. Hydrophobic groups can preferentially interact with each other rather than with water molecules (figure 5.9c). Such hydrophobic interactions very often play a major role in antibody-antigen binding.

4 Van der Waals. The forces between molecules which depend upon interaction between the external 'electron clouds' (figure 5.9d). Van der Waals forces account for a minority, between 2% and 20%, of the total interaction energy in the binding of a typical antibody to its antigen.

One essential feature common to all four types of force is that they depend upon the close approach of both molecules before the forces become of significant magnitude, the more so if water molecules are excluded. And this is at the heart of the combination of antigen and antibody. The **complementary** electron-cloud shapes on the combining site of the antibody and the surface determinant of the antigen enable the two molecules to fit snugly together (cf. figure 5.5) so that the **intermolecular distance becomes very small** and the **noncovalent protein interaction forces are considerably increased**; the greater the areas of antigen and antibody which fit together, the greater the force of attraction, particularly if there is apposition of opposite charges and hydrophobic groupings.

By contrast, when the electron clouds of the two molecules effectively overlap, powerful repulsive forces are generated and energy must be expended in displacing the overlapping residues from their normal equilibrium positions.

AFFINITY MEASURES STRENGTH OF BINDING OF ANTIGEN AND ANTIBODY

We saw from the electron microscope studies on the interaction between a divalent DNP conjugate and antibody that each DNP group fitted into one antibody

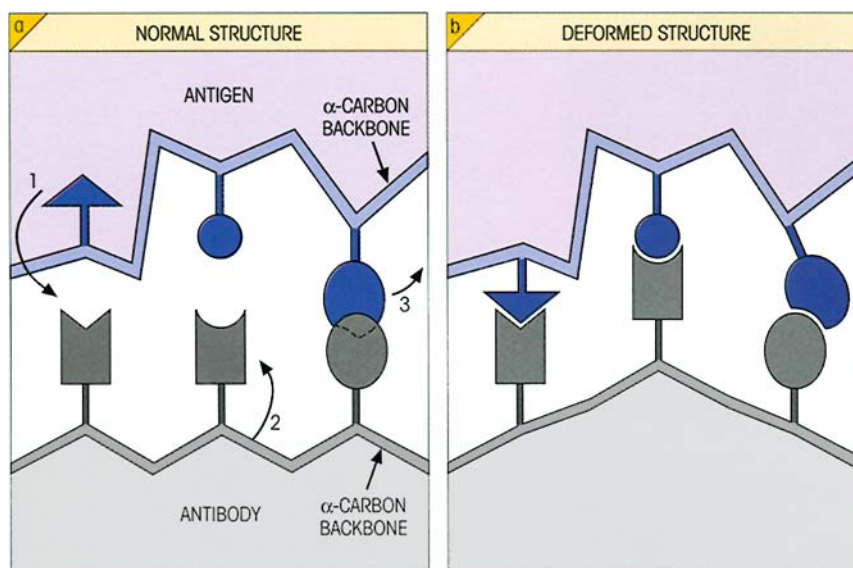


Figure 5.7. Flexibility of antigen and antibody contributes to binding affinity. In the illustration we show: (a) three types of energy-consuming reactions required to deform both molecules, thereby allowing (b) positive interaction between residues in the paratope and epitope — (1) bond rotation exposing a hydrophobic side-chain normally buried within the interior, (2) flexing of the α -carbon backbone to bring interacting residues close together, and (3) lateral displacement of a residue whose electron clouds would overlap with those of its opposite partner. Provided that the total deformation energy is less than the energy of attraction of the deformed molecules, complex formation will be favored. Expressed mathematically:

$$\text{Free energy of deformation} = (\Delta G_1 + \Delta G_2 + \Delta G_3) = \Delta G_{\text{def}}$$

where ΔG_1 is the Gibbs free energy change of reaction (1) and so on.

Energy of association of deformed antigen and antibody = the sum of the individual binding energies = $\Delta G_{\text{binding}}$ (an attractive force gives a negative ΔG).

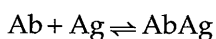
Overall energy changes for complex formation

$$= \Delta G_{\text{def}} + \Delta G_{\text{binding}} = -RT \ln K_a$$

Provided that $-\Delta G_{\text{binding}} > \Delta G_{\text{def}}$, antigen and antibody will associate at equilibrium with a reasonable association affinity constant K_a . (R , gas constant; T , absolute temperature; $\ln$, natural logarithm.)

If the antigen were completely rigid, it would be unable to approach close enough to the antibody to generate significant binding energy.

combining site (figures 3.1 and 3.2). This means that small haptens by themselves are monovalent with respect to reaction with antibody. The experiment on mixing hapten with antibody in a dialysis bag (figure 5.8) showed that the combination with antibody was reversible and that the complex so formed could readily dissociate depending upon the strength of binding, which we call **affinity**. In the simplistic situation of one Fab arm binding in isolation to one epitope on the antigen, the binding strength can be defined through the **equilibrium constant (K_a)** of the **association reaction**:



given by the mass action equation

$$K_a = \frac{[\text{AbAg}]}{[\text{Ab}][\text{Ag}]}$$

where $[\text{Ab}]$ is the concentration of free antibody combining sites and $[\text{Ag}]$ is the concentration of free antigen at equilibrium. If the antibody and antigen fit together very closely, the equilibrium will lie well over to the right of the association reaction; we refer to such antibodies which bind strongly to the antigen as **high affinity antibodies**. At a certain free antigen concentration $[\text{Ag}_c]$ where half of the antibody sites are bound:

$$[\text{AbAg}] = [\text{Ab}] \text{ and } K_a = 1/[\text{Ag}_c]$$

i.e. the affinity constant K_a is equal to the reciprocal of the concentration of free antigen at the equilibrium point where half the antibody sites are in the bound form. In other words, when an antibody has a high affinity constant and binds antigen strongly, it only needs a low antigen concentration to half-saturate the antibody. An individual epitope on the surface of a complex antigen is by definition monovalent, and the

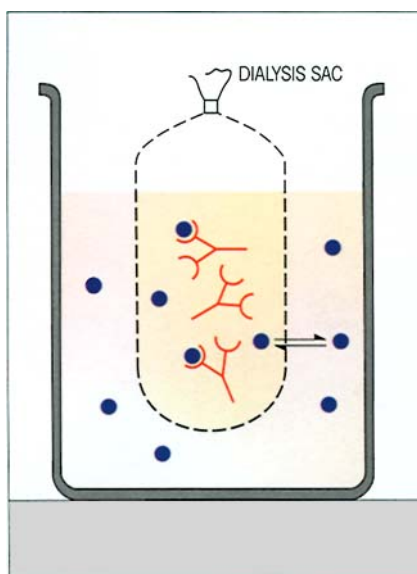
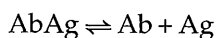


Figure 5.8. Reversibility of the interaction between antibody (—Y) and hapten (●). Within the dialysis sac the hapten is partly in the free form and partly bound to antibody according to the affinity of the antibody. Only hapten can diffuse through the dialysis membrane, and the external concentration then will equal the concentration of unbound hapten within the sac. Measurement of total hapten in the dialysis sac then enables the amount bound to antibody to be calculated. Constant renewal of the external buffer will lead to total dissociation and loss of hapten from inside the dialysis sac showing the reversible nature of the antigen–antibody bond.

strength of its combination with one antigen-binding arm (Fab) of an antibody is therefore defined by an affinity constant.

Affinity can equally well be formulated in terms of the **dissociation constant** (K_d) of the reaction:



Expressing concentrations in moles per liter:

$$K_d = \frac{[\text{Ab moles/l}][\text{Ag moles/l}]}{[\text{AbAg moles/l}]}$$

Clearly, K_d is the reciprocal of K_a , i.e. $1/K_a$, and has the units moles/l or M. Conversely, K_a is expressed in the units 1/mole or M^{-1} and has the advantage that the stronger the binding, the higher the number.

The value of K_a is determined by the difference in free energy (ΔG) between the antigen and antibody in the free state on the one hand and in the complexed form on the other, according to the equation:

$$\Delta G = -RT \ln K_a$$

where R is the universal gas constant, T is the absolute temperature and $\ln$ is the natural logarithm.

One can study the interaction of hapten and antibody by the dialysis method described in figure 5.8 and use the data to calculate the affinity constant from the mass action equation (figure 5.10). K_a values may sometimes be as high as 10^{12}M^{-1} .

Analysis of the binding at different hapten concentrations generally shows a heterogeneity (figure 5.10) which indicates that most antisera, even those raised against haptens with a simple structure, contain a variety of different antibodies with a range of binding affinities which depend upon the area of contact between the antibody and the hapten or epitope, the closeness of fit, conformational changes necessitated by electron-cloud overlap and the distribution of charged and hydrophobic groups.

The avidity of antiserum for antigen — the bonus effect of multivalency

While the term affinity describes the binding of antibody to a monovalent hapten or single antigen determinant, in most practical circumstances we are concerned with the interaction of an antiserum (i.e. the serum from an immunized individual) with a multivalent antigen. The term employed to express this binding is **avidity** or **functional affinity**.

The factors which contribute to avidity are complicated, including as they do the heterogeneity of antibodies in a given serum which are directed against each determinant on the antigen, and the heterogeneity of the determinants themselves (figures 5.4 and 5.10). But yet a further factor must be considered. The multivalency of most antigens leads to an interesting **bonus** effect in which the binding of antigen to antibody by multiple links is always greater, usually many-fold greater, than the arithmetic sum of the individual antibody bonds. This is illustrated in figure 5.11. The mechanism of this effect may be interpreted by considering an analogy. Let us fabricate an unheard of disease in which we cannot stop our hands opening and closing continuously. If we now try to hold an object in one hand, it will fall the moment we open that hand. However, if we use both hands to hold the object, provided that we open and close our hands at different times, there is much less chance of the object falling. The reversible combination of antigen and antibody is like the opening and closing of the hands; the more valencies holding the antigen, the less likely it is to be lost when the complex dissociates at any one binding site (figure 5.12).

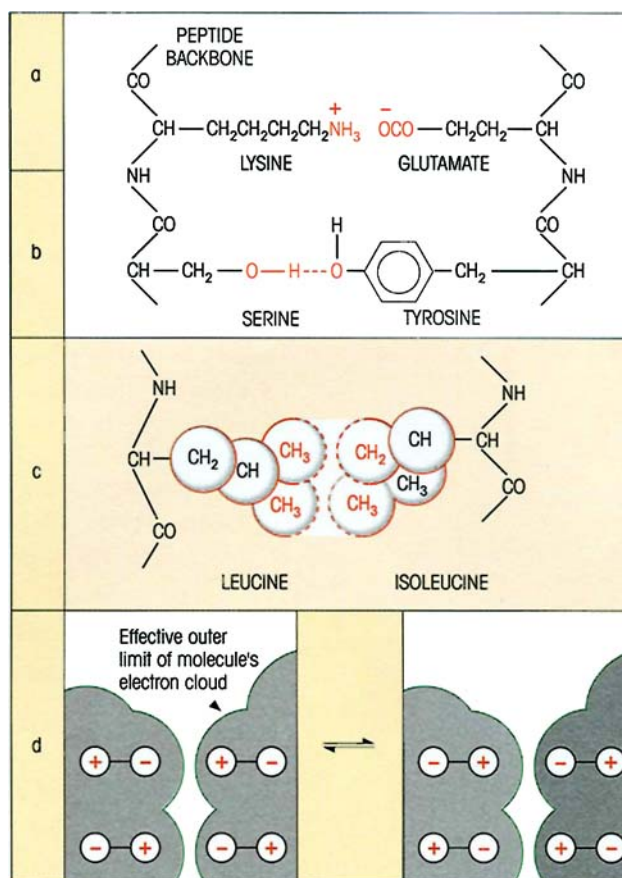


Figure 5.9. Protein-protein interactions. (a) Coulombic attraction between oppositely charged ionic groups on the two protein side-chains as illustrated by an ionized amino group (NH_3^+) on a lysine of one protein and an ionized carboxyl group (COO^-) of glutamate on the other. The force of attraction is inversely proportional to the square of the distance between the charges. Thus, as the charges come closer together, the attractive force increases considerably: if we halve the distance apart, we quadruple the attraction. Furthermore, since the dielectric constant of water is extremely high, the exclusion of water molecules through the contiguity of the interacting residues would greatly increase the force of attraction. Dipoles on antigen and antibody can also attract each other. In addition, electrostatic forces may be generated by charge transfer reactions between antibody and antigen; for example, an electron-donating protein residue such as tryptophan could part with an electron to a group such as dinitrophenyl (DNP) which is electron accepting, thereby creating an effective +1 charge on the antibody and -1 on the antigen. (b) Hydrogen bonding between two proteins involving the formation of reversible hydrogen bridges between hydrophilic groups, such as OH , NH_2 and COOH , depends very much upon the close approach of the two molecules carrying these groups. Although H bonds are relatively weak, because they are essentially electrostatic in nature, exclusion of water between the reacting side-chains would greatly enhance the binding energy through the gross reduction in dielectric constant. (c) Nonpolar, hydrophobic groups such as the side-chains of valine, leucine and isoleucine tend to associate in an

aqueous environment. The driving force for this hydrophobic interaction derives from the fact that water in contact with hydrophobic molecules (—), with which it cannot H bond, will associate with other water molecules, but the number of configurations which allow H bonds to form will not be as great as that occurring when they are surrounded completely by other water molecules, i.e. the entropy is lower. The greater the area of contact between water and hydrophobic surfaces, the lower the entropy and the higher the energy state. Thus, if hydrophobic groups on two proteins come together so as to exclude water molecules between them (— —), the net surface in contact with water is reduced and the proteins take up a lower energy state than when they are separated (in other words, there is a force of attraction between them). (d) Van der Waals force: the interaction between the electrons in the external orbitals of two different macromolecules may be envisaged (for simplicity!) as the attraction between induced oscillating dipoles in the two electron clouds. The nature of this interaction is difficult to describe in non-mathematical terms, but it has been likened to a temporary perturbation of electrons in one molecule effectively forming a dipole which induces a dipolar perturbation in the other molecule, the two dipoles then having a force of attraction between them; as the displaced electrons swing back through the equilibrium position and beyond, the dipoles oscillate. The force of attraction is inversely proportional to the seventh power of the distance and, as a result, this rises very rapidly as the interacting molecules come closer together.

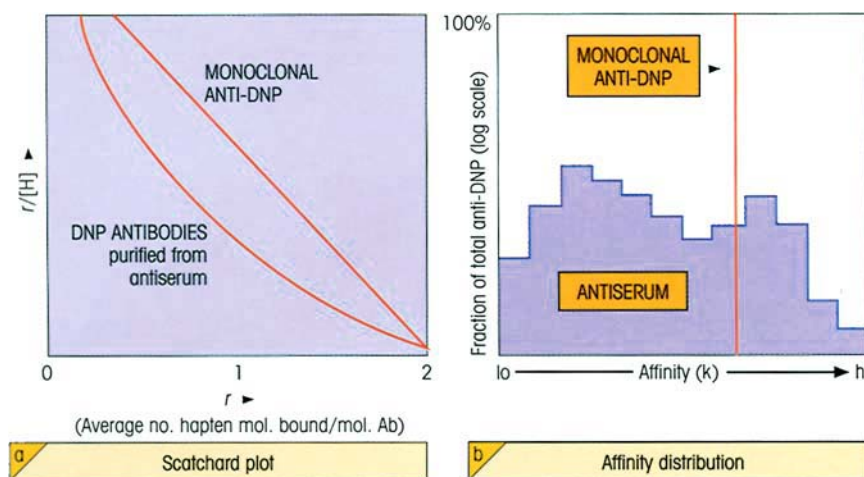


Figure 5.10. Heterogeneity of IgG antihapten (dinitrophenyl, DNP) antibodies from the serum of an immunized animal contrasting with the homogeneity of a monoclonal IgG anti-DNP. (a) A Scatchard plot of hapten binding to antibodies purified from the serum. If r represents the average number of DNP molecules bound to each antibody molecule, of affinity constant k and number of binding sites n , in the presence of a free hapten concentration $[H]$, then from the mass action equation of equilibrium relationships (p. 86) it can be shown that:

$$r/[H] = nk - rk$$

Thus, a Scatchard plot of $r/[H]$ against r for a single antibody species will be a straight line of slope $-k$ as seen for the monoclonal antibody; the deviation from a straight line given by anti-DNP from the antiserum clearly indicates the existence of antibodies with different affinities, as may be confirmed by the binding of labeled DNP to

many different bands after separation of the individual antibodies by isoelectric focusing of the serum. Extrapolation to $r/[H]=0$ (at infinitely high concentration of antigen) gives the number of binding sites on each IgG molecule as two (cf. figure 3.2). For IgM antibodies, the value would be 10. Because the slope of the Scatchard plot varies with $r/2$ (the fractional occupancy of the antibody combining sites for a bivalent Ab), the affinity (which $= -\text{slope}$) will vary depending upon the range of values of r utilized in the experiments: thus affinity must be defined in terms of standard conditions of antibody dilution and concentration of hapten. (b) Histogram showing a typical distribution of antibody affinities in the anti-DNP serum. Measurable affinities tend to range between 10^4 and 10^{10} or 10^{11} M^{-1} and have skewed and not necessarily unimodal distributions. The monoclonal antibody of course gives a single affinity value since it is a homogeneous protein.

The same considerations apply to the binding of antibody to a polymeric antigen with repeating determinants, such as ovalbumin substituted by several DNP groups, or most bacterial polysaccharides or red cells with repeating blood-group determinants. As one moves from a univalent Fab fragment to a divalent IgG to a pentameric IgM, the bonus effect of multivalency produces striking increases in the strength of antigen-antibody complex formation.

Avidity, being a measure of the functional affinity of an antiserum for the whole antigen, is of obvious relevance to the reaction with antigen in the body. High avidity is superior to low for a wide variety of functions *in vivo*, e.g. immune elimination of antigen, virus neutralization, protective role against bacteria, and so on.

THE SPECIFICITY OF ANTIGEN RECOGNITION BY ANTIBODY IS NOT ABSOLUTE

The ability of antibodies to discriminate between different antigens was well illustrated by the range of

reactivity of an antihapten for a series of structurally related molecules as described in table 5.1. Since the strength of the reaction can be quantified by the affinity or avidity, we would relate the **specificity** of an antiserum to its relative avidity for the antigens which are being discriminated.

In so far as we recognize that an antiserum may have a relatively greater avidity for one antigen rather than another, by the same token we are saying that the antiserum is displaying relative rather than absolute specificity; in practice we speak of degrees of **cross-reactivity**. An antiserum raised against a given antigen can cross-react with a partially related antigen which bears one or more identical or similar determinants. In figure 5.13 it can be seen that an antiserum to antigen₁ (Ag_1) will react less strongly with Ag_2 , which bears just one identical determinant, because only certain of the antibodies in the serum can bind. Ag_3 , which possesses a similar but not identical determinant, will not fit as well with the antibody and the binding is even weaker. Ag_4 , which has no structural similarity at all, will not react significantly with the antibody. Thus, based upon stereochemical considerations, we can see why the

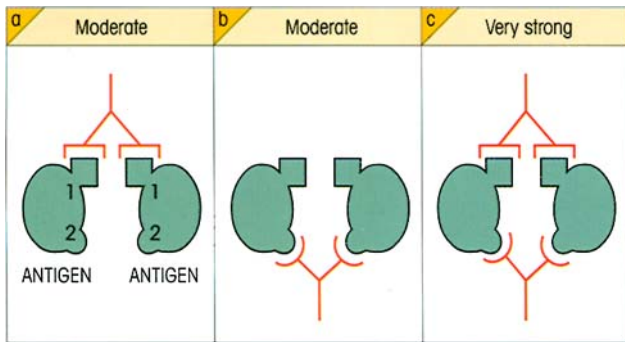


Figure 5.11. The 'bonus' effect of multivalent attachment on binding strength. The force binding the two antigen molecules in (c) with two antibody bridges is many-fold greater than (a) + (b). Thus, the binding of the two antibodies to determinants 1 and 2 on the antigen can each be described by the individual change in free energy state on forming the epitope–paratope bond with its corresponding affinity constant, i.e.:

$$\Delta G_1 = -RT \ln k_1 \text{ and } \Delta G_2 = -RT \ln k_2$$

For both antibodies operating in conjunction, the overall free energy change giving the avidity (K_{avid}) would be:

$$\Delta G = \Delta G_1 + \Delta G_2 = -RT \ln k_1 - RT \ln k_2 \\ = -RT(\ln k_1 + \ln k_2) = -RT(\ln k_1 \times k_2)$$

$$\text{Since } \Delta G = -RT \ln K_{\text{avid}}, K_{\text{avid}} = k_1 \times k_2$$

The tremendous increase in equilibrium constant resulting from **multiplying the contributing affinities** is responsible for the bonus effect. To give a concrete example, if k_1 is 10^4 and k_2 is 10^3 , K_{avid} would be 10^7 M^{-1} . In practice, this bonus effect would be reduced by entropy losses resulting from any restriction of the flexibility of the antibody molecules required by adaptation to the spatial demands of the epitopes and from the restriction in translational movement of the individual components.

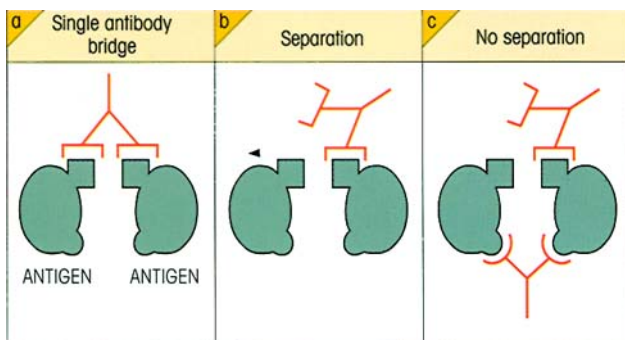


Figure 5.12. The mechanism of the bonus effect. Each antigen–antibody bond is reversible and, with a single antibody bridge between two antigen molecules (a), dissociation of either bond could enable an antigen molecule to 'escape' as in (b). If there are two antibody bridges, even when one dissociates the other prevents the antigen molecule from escaping and holds it in position ready to reform the broken bond (c). In effect, the orientation of the broken bond greatly increases the effective combining concentration of antibody and thereby speeds up the velocity of the association reaction: $V_a = k_a[\text{Ag}][\text{Ab}]$.

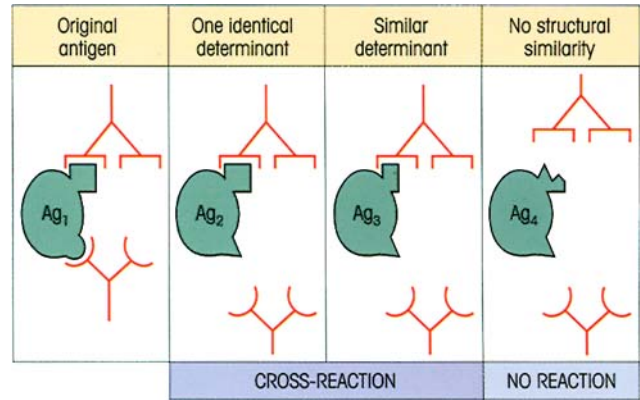


Figure 5.13. Specificity and cross-reaction. The avidity of the serum antibodies (H and O) for $\text{Ag}_1 > \text{Ag}_2 > \text{Ag}_3 \gg \text{Ag}_4$.

avidity of the antiserum for Ag_2 and Ag_3 is less than for the homologous antigen, while for the unrelated Ag_4 it is negligible. It would be customary to describe the antiserum as being highly specific for Ag_1 in relation to Ag_4 , but cross-reacting with Ag_2 and Ag_3 to different extents.

By being directed towards single epitopes on the antigen, monoclonal antibodies frequently show high specificity in terms of their low cross-reactivity with other antigens. Occasionally, however, one sees quite unexpected binding to antigens which react poorly, if at all, with a specific antiserum. It is an instructive exercise to see how it is that a polyclonal antiserum containing a heterogeneous collection of antibodies can be more specific in discriminating between two antigens than can a monoclonal antibody. The six hypervariable regions of an antibody encompass a relatively large molecular area composed of highly diverse amino acid side-chains and it is self evident that a number of different epitopic structures could fit into different parts of this hypervariable 'terrain', albeit with a spectrum of combining affinities.

Thus, each antibody will react not only with the antigen which stimulated its production, but also with some possibly quite unrelated molecules. Figure 5.14 explains (we hope) how this may translate into a higher specificity for the polyclonal serum.

WHAT THE T-CELL SEES

We have on several occasions alluded to the fact that the T-cell receptor sees antigen on the surface of cells associated with an MHC class I or II molecule. Now is the time for us to go into the nuts and bolts of this relationship.

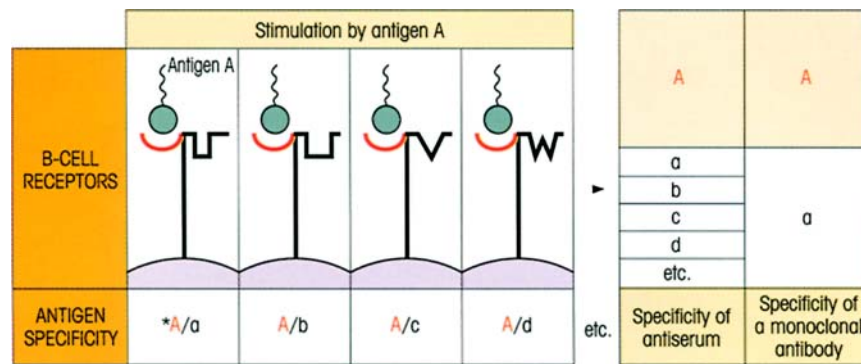


Figure 5.14. The specificity of an antiserum derives from the reactivity common to the component antibodies. Antigen A stimulates lymphocytes whose polyfunctional receptors bind A but could also bind other determinants as indicated (small letters). All antibodies in the resulting antiserum will have anti-A as a common specificity, but the other specificities will be so diverse that none of them will reach

appreciable concentrations to cross-react significantly with another antigen bearing a or b, etc., i.e. the antiserum shows specificity for A. On the other hand, a monoclonal antibody cannot dilute out its alternative specificity and so in the example shown with an asterisk there would be strong cross-reaction with an unrelated antigen a. (With acknowledgment to Talmage D.W. (1959) *Science* **129**, 1643.)

Haplotype restriction reveals the need for MHC participation

It has been established in tablets of stone that T-cells bearing $\alpha\beta$ receptors, with some exceptions (cf. p. 100), only respond when the antigen-presenting cells express the same MHC haplotype as the host from which the T-cells were derived (Milestone 5.1). This **haplotype restriction** on T-cell recognition tells us unequivocally that MHC molecules are intimately and necessarily involved in the interaction of the antigen-bearing cell with its corresponding antigen-specific T-lymphocyte. We also learn that, generally speaking, cytotoxic T-cells recognize antigen in the context of class I MHC, and helper T-cells which interact with macrophages respond when the antigen is associated with class II molecules.

Accepting then the participation of MHC in T-cell recognition, what of the antigen? For some time it was perplexing that, in so many systems, antibodies raised to the native antigen failed to block cytotoxicity (cf. figure M5.1.1b), despite consistent success with anti-MHC class I sera. We now know why.

T-cells recognize a linear peptide sequence from the antigen

In Milestone 5.1, we commented on experiments involving influenza nucleoprotein-specific T-cells which could kill cells infected with influenza virus. Killing occurs after the cytotoxic T-cell adheres strongly to its target through recognition of specific surface molecules. It is curious then that the nucleoprotein, which lacks a signal sequence or transmembrane region and

so cannot be expressed on the cell surface, can nonetheless function as a target for cytotoxic T-cells, particularly since we have already noted that antibodies to native nucleoprotein have no influence on the killing reaction (figure M5.1.1b). Furthermore, uninfected cells do not become targets for the cytotoxic T-cells when whole nucleoprotein is added to the culture system. However, if, instead, we add a series of short peptides with sequences derived from the primary structure of the nucleoprotein, the uninfected cells now become susceptible to cytolytic T-cell attack (figure 5.15).

Thus was the secret revealed! The startling reality is that T-cells recognize linear peptides derived from the antigen, and that is why antibodies raised against nucleoprotein in its native three-dimensional conformation (cf. figure 5.2) do not inhibit killing. Note that only certain nucleoprotein peptides were recognized by the polyclonal T-cells in the donor population and these are to be regarded as T-cell epitopes. When clones of identical specificity are derived from these T-cells, each clone reacts with only one of the peptides; in other words, like B-cell clones, each clone is specific for one corresponding epitope.

Entirely analogous results are obtained when T-helper clones are stimulated by antigen-presenting cells to which certain peptides derived from the original antigen have been added. Again, by synthesizing a series of such peptides, the T-cell epitope can be mapped with some precision.

The conclusion is that the **T-cell recognizes both MHC and peptide** and we now know that the peptide which acts as a T-cell epitope lies along the groove formed by the α -helices and the β -sheet floor of the

Milestone 5.1 — MHC Restriction of T-cell Reactivity

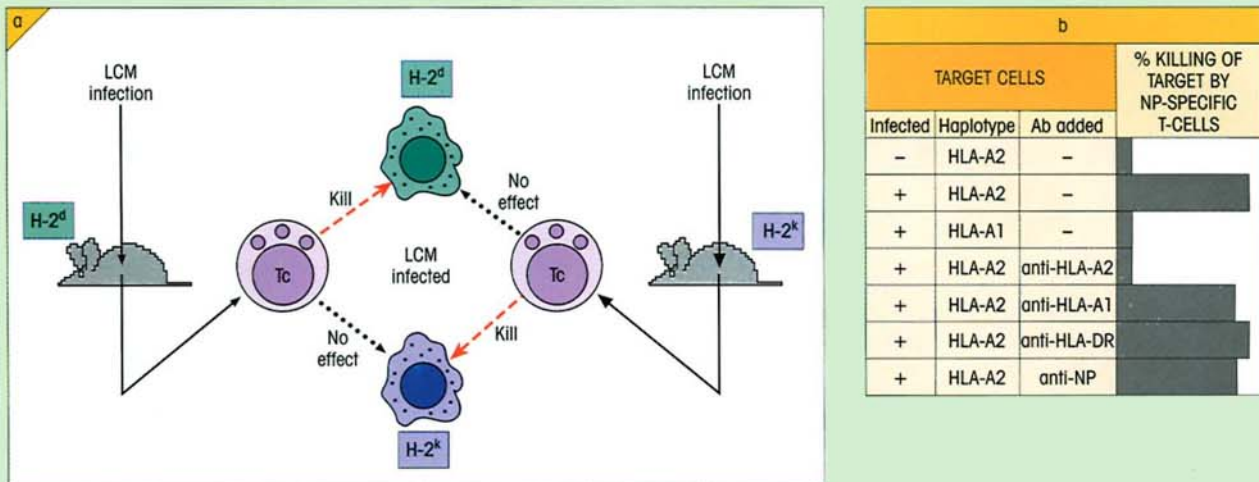


Figure M5.1.1. T-cell killing is restricted by the MHC haplotype of the virus-infected target cells. (a) Haplotype-restricted killing of lymphocytic choriomeningitis (LCM) virus-infected target cells by cytotoxic T-cells. Killer cells from H-2^d hosts only killed H-2^d-infected targets, not those of H-2^k haplotype and vice versa.

(b) Killing of influenza-infected target cells by influenza nucleoprotein (NP)-specific T-cells from an HLA-A2 donor (cf. p. 356 for human MHC nomenclature). Killing was restricted to HLA-A2 targets and only inhibited by antibodies to A2, not to A1, nor to the class II HLA-DR framework or native NP antigen.

The realization that the MHC, which had figured for so long as a dominant controlling element in tissue graft rejection, should come to occupy the center stage in T-cell reactions has been a source of fascination and great pleasure to immunologists—almost as though a great universal plan had been slowly unfolding.

One of the seminal observations which helped to elevate the MHC to this lordly position was the dramatic Nobel prize-winning revelation by Doherty and Zinkernagel that cytotoxic T-cells taken from an individual recovering from a viral infection will only kill virally infected cells which share an MHC haplotype with the host. They found that cytotoxic T-cells from mice of the H-2^d haplotype infected with lymphocytic choriomeningitis virus could kill virally infected cells derived from any H-2^d strain but not cells of H-2^k or other H-2 haplotype. The reciprocal experiment with H-2^k mice shows that this is not just a special property associated with H-2^d (figure M5.1.1a). Studies with recombinant strains (cf. table 4.2) pinpointed class I MHC as the restricting element and this was confirmed by showing that antibodies to class I MHC block the killing reaction.

The same phenomenon has been repeatedly observed in the human. HLA-A2 individuals recovering from influenza have cytolytic T-cells which kill HLA-A2 target cells infected with influenza virus, but not cells of a different HLA-A tissue-type specificity (figure M5.1.1b). Note how

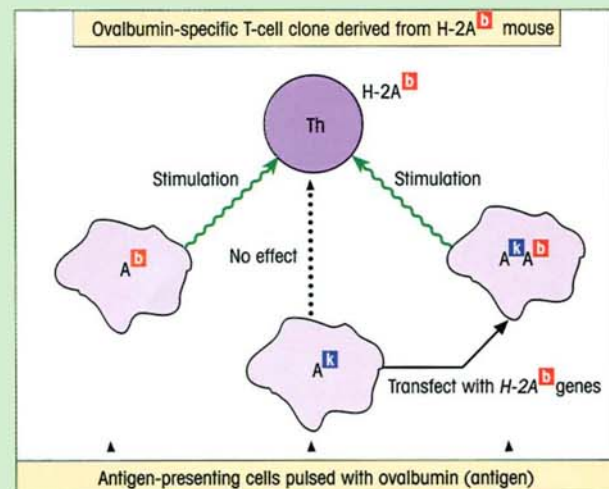


Figure M5.1.2. The T-cell clone only responds by proliferation *in vitro* when the antigen-presenting cells (e.g. macrophages) pulsed with ovalbumin express the same class II MHC.

cytotoxicity could be inhibited by antiserum specific for the donor HLA-A type, but not by antisera to the allelic form HLA-A1 or the HLA-DR class II framework. Of striking significance is the inability of antibodies to the nucleoprotein to block T-cell recognition even though the T-cell specificity in these studies was known to be directed towards this antigen. Since the antibodies react with

(continued)